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IMAGE FORMING APPARATUS AND TONER REPLENISHMENT CONTROL METHOD IN IMAGE FORMING APPARATUS

Abstract

According to the image forming apparatus according to the present disclosure, when at least one of accumulation necessary replenishment times of two types of toner becomes equal to or greater than a toner replenishment threshold, replenishment of the one toner is performed. The replenishment of this one toner is performed for the accumulation necessary replenishment time of the one toner with a continuous replenishment allowable time as an upper limit. Thereafter, according to the accumulation necessary replenishment time of the other toner and the current accumulation necessary replenishment time of the one toner, the replenishment of the other toner is performed, the replenishment of the one toner is resumed, or the replenishment of both toners is brought into a stop state.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an image forming apparatus and a toner replenishment control method in the image forming apparatus, and particularly relates to an image forming apparatus and a toner replenishment control method in the image forming apparatus that perform color image formation processing by an electrophotographic method.

BACKGROUND ART

[0002] This type of image forming apparatus is generally provided with four development apparatuses responsible for development of four colors of yellow (Y), magenta (M), cyan (C), and black (K). When development is performed by each development apparatus, toner is consumed, but each development apparatus is replenished with the toner in order to compensate for consumption of this toner. The replenishment of this toner is performed by a toner replenishment apparatus, and there is a toner replenishment apparatus having a configuration in which toner replenishment of two colors of four colors is performed by one (common) drive source, and toner replenishment of the other two colors is performed by another drive source. According to this configuration, since the number of drive sources is reduced, cost reduction of the entire image forming apparatus including the toner replenishment apparatus is achieved. On the other hand, since each drive source selectively performs toner replenishment of two colors, it is not possible to simultaneously perform toner replenishment of the two colors. Therefore, it is important for each drive source to perform toner replenishment of two colors performed by the drive source in good balance.

[0003] There is a technique in which the amount of a first developer to be supplied to a first development apparatus is calculated, the amount of a second developer to be supplied to a second development apparatus is calculated, a first drive time for driving a first developer supply member responsible for supply of the first developer and a second drive time for driving a second developer supply member responsible for supply of the second developer are allocated according to the ratio between the amount of the first developer and the amount of the second developer that are calculated, and the first developer supply member and the second developer supply member are selectively driven according to this allocated time. This technique reduces the density variation of the developer is reduced, and reduces the disadvantage that, for example, the density of the developer of one of the first development apparatus and the second development apparatus increases and the density of the developer of the other becomes insufficient.

SUMMARY

Technical Problem

[0004] In the technique described above, the first drive time for driving the first developer supply member for supply of the first developer and the second drive time for driving the second developer supply member for supply of the second developer are allocated according to the ratio between the amount of the first developer to be supplied to the first development apparatus and the amount of the second developer to be supplied to the second development apparatus. That is, at least processing (calculation) that obtains the ratio and processing that allocates the first drive time and the second drive time according to the ratio are performed. The replenishment manner of the toner including these processing is accordingly complicated, and it is necessary to have a means,

such as a CPU responsible for the processing, that has a corresponding ability (performance). This leads to an increase in cost (increase in price) of the entire image forming apparatus, in other words, hinders a reduction in cost of the entire image forming apparatus due to a reduction in the number of drive sources.

[0005] Therefore, an object of the present disclosure is to provide a novel technique that can perform two types of toner replenishment in good balance with an extremely simple and uncomplicated manner in an image forming apparatus having a configuration in which two types of toner replenishment are performed by one drive source and a toner replenishment control method in the image forming apparatus.

Solution to Problem

[0006] In order to achieve this object, the present disclosure includes a first disclosure relating to an image forming apparatus and a second disclosure relating to a toner replenishment control method in the image forming apparatus.

[0007] The first disclosure is an image forming apparatus that performs image formation processing by an electrophotographic method, and includes a first accommodation, a first replenishment member, a second accommodation, a second replenishment member, a drive source, a drive controller, a derivator, and an accumulator. The first accommodation accommodates a first toner. The first replenishment member replenishes the first toner accommodated in the first accommodation to a first development unit. The second accommodation accommodates second toner. The second replenishment member replenishes the second toner accommodated in the second accommodation to a second development unit. The drive source selectively drives the first replenishment member and the second replenishment member. The drive controller controls the drive source. The derivator derives a first necessary replenishment time and a second necessary replenishment time based on image data to be subjected to the image formation processing. The first necessary replenishment time is a replenishment time of the first toner necessary for compensating for consumption of the first toner consumed by the first development unit. The second necessary replenishment time is a replenishment time of the second toner necessary for compensating for the consumption of the second toner consumed by the second development unit. The accumulator accumulates a shortage of an actual replenishment time of the first toner by the first replenishment member with respect to the first necessary replenishment time every time the image formation processing is performed and obtains a first accumulation time which is an accumulation value of the shortage. Also, the accumulator accumulates a shortage of an actual replenishment time of the second toner by the second replenishment member with respect to the second necessary replenishment time every time the image formation processing is performed and obtains a second accumulation time which is an accumulation value of the shortage.

[0008] Here, when at least one of the first accumulation time and the second accumulation time is equal to or greater than a predetermined first threshold, the drive controller specifies the first accumulation time or the second accumulation time that is equal to or greater than the first threshold as one accumulation time, and specifies the first replenishment member or the second replenishment member related to the one accumulation time as one replenishment member. The drive controller drives the one replenishment member for the one accumulation time with a predetermined second threshold as an upper limit, and then stops drive of the one replenishment member. Furthermore, according to the other accumulation time, which is not the one accumulation time, of the first accumulation time and the second accumulation time and the current one accumulation time, the drive controller controls the drive source so as to drive the other replenishment member, which is not the one replenishment member, of the first replenishment member and the second replenishment member, or drives the one replenishment member again, or bring the one replenishment member and the other replenishment member into a stop state without driving both of the one replenishment member and the other replenishment member. In particular, when the other replenishment member is driven, the other replenishment member is driven for the

other accumulation time with the second threshold as an upper limit. When the one replenishment member is driven again, the one replenishment member is driven for the current one accumulation time with the second threshold as an upper limit.

[0009] When both the first accumulation time and the second accumulation time are equal to or greater than the first threshold, the drive controller specifies the longer one of the first accumulation time and the second accumulation time as the one accumulation time.

[0010] In the present first disclosure, a first toner density detector and a second toner density detector may be further included. The first toner density detector detects the density of the first toner contained in a developer in the first development unit, and the second toner density detector detects the density of the second toner contained in a developer in the second development unit. When both the first accumulation time and the second accumulation time are equal to or greater than the first threshold and when the first accumulation time and the second accumulation time have a length equal to each other, the drive controller specifies, as the one accumulation time, the first accumulation time or the second accumulation time corresponding to a lower one of a density detection value by the first toner density detector and a density detection value by the second toner density detector.

[0011] The second threshold in the present first disclosure is desirably equal to or greater than the first necessary replenishment time in a case where a printing rate related to the first toner based on the image data is 50% and equal to or less than the first necessary replenishment time in a case where the printing rate is 90%. In other words, the second threshold is desirably a value equal to or greater than the second necessary replenishment time in a case where the printing rate related to the second toner based on the image data is 50% and equal to or less than the second necessary replenishment time in a case where the printing rate is 90%.

[0012] In addition, when at least one of the first accumulation time and the second accumulation time is equal to or greater than a predetermined third threshold, the drive controller desirably controls the drive source so as to interrupt the image formation processing, drive the first replenishment member for the first accumulation time, and drive the second replenishment member for the second accumulation time, that is, execute forced replenishment. The third threshold is a value larger (corresponding to a long time) than the first threshold and the second threshold.

[0013] The second disclosure of the present disclosure is a toner replenishment control method in an image forming apparatus that performs image formation processing by an electrophotographic method, and includes a drive control step, a derivation step, and an accumulation step. Here, the image forming apparatus includes the first accommodation, the first replenishment member, the second accommodation, the second replenishment member, and the drive source. The first accommodation accommodates the first toner. The first replenishment member replenishes the first toner accommodated in the first accommodation to the first development unit. The second accommodation accommodates the second toner. The second replenishment member replenishes the second toner accommodated in the second accommodation to the second development unit. The drive source selectively drives the first replenishment member and the second replenishment member. Then, the drive control step includes controlling the drive source. The derivation step includes deriving the first necessary replenishment time and the second necessary replenishment time based on the image data to be subjected to the image formation processing. The first necessary replenishment time is a replenishment time of the first toner necessary for compensating for consumption of the first toner consumed by the first development unit. The second necessary replenishment time is a replenishment time of the second toner necessary for compensating for the consumption of the second toner consumed by the second development unit. Furthermore, the accumulation step includes accumulating a shortage of an actual replenishment time of the first toner by the first replenishment member with respect to the first necessary replenishment time every time the image formation processing is performed and obtaining a first accumulation time, which is an accumulation value of the shortage. Also, the accumulation step includes accumulating

a shortage of an actual replenishment time of the second toner by the second replenishment member with respect to the second necessary replenishment time every time the image formation processing is performed and obtaining a second accumulation time which is an accumulation value of the shortage.

[0014] The drive control step includes, when at least one of the first accumulation time and the second accumulation time is equal to or greater than a predetermined first threshold, specifying the first accumulation time or the second accumulation time that is equal to or greater than the first threshold as one accumulation time, and specifying the first replenishment member or the second replenishment member related to the one accumulation time as one replenishment member. The drive control step includes driving the one replenishment member for the one accumulation time with a predetermined second threshold as an upper limit, and then stopping drive of the one replenishment member. Furthermore, the drive control step includes, according to the other accumulation time, which is not the one accumulation time, of the first accumulation time and the second accumulation time and the current one accumulation time, controlling the drive source so as to drive the other replenishment member, which is not the one replenishment member, of the first replenishment member and the second replenishment member, or driving the one replenishment member again, or bringing the one replenishment member and the other replenishment member into a stop state without driving both of the one replenishment member and the other replenishment member. In particular, when the other replenishment member is driven, the other replenishment member is driven for the other accumulation time with the second threshold as an upper limit. When the one replenishment member is driven again, the one replenishment member is driven for the current one accumulation time with the second threshold as an upper limit.

Advantageous Effects of Disclosure

[0015] According to the present disclosure, it is possible to replenish two types of toner of the first toner and the second toner in good balance with an extremely simple and uncomplicated manner as compared with the technique disclosed in Patent Document 1 described above. This is extremely advantageous for an image forming apparatus having a configuration of achieving cost reduction by replenishing two types of toner by one drive source.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0016] FIG. 1 is a view schematically illustrating an internal configuration of an image forming apparatus according to a first example of the present disclosure.

[0017] FIG. 2 is a perspective view of a toner replenishment apparatus in the first example.

[0018] FIG. 3 is a perspective cross-sectional view of the toner replenishment apparatus in the first example.

[0019] FIG. 4 is a view that describes an operation of the toner replenishment apparatus in the first example.

[0020] FIG. 5 is a view illustrating an example of a relationship between a printing timing and a toner replenishment timing in the first example.

[0021] FIG. 6 is a view that describes an example of a toner replenishment manner in the first example.

[0022] FIG. 7 is a view that describes another example of the toner replenishment manner in the first example.

[0023] FIG. 8 is a block diagram illustrating an electrical configuration of the image forming apparatus according to the first example.

[0024] FIG. 9 is a flowchart showing a flow of a toner replenishment control task in the first example.

[0025] FIG. **10** is a flowchart showing a flow of a part of toner replenishment processing in the first example.

[0026] FIG. **11** is a flowchart showing a flow of another part of the toner replenishment processing in the first example.

[0027] FIG. **12** is a flowchart showing a flow of the remaining part of the toner replenishment processing in the first example.

[0028] FIG. **13** is a view conceptually illustrating a configuration of a forced replenishment threshold table in a second example of the present disclosure.

[0029] FIG. **14** is a view conceptually illustrating a configuration of a threshold coefficient table according to a third example of the present disclosure.

DESCRIPTION OF EMBODIMENTS

First Example

[0030] The first example of the present disclosure will be described with an image forming apparatus **10** illustrated in FIG. **1** as an example.

[0031] The image forming apparatus **10** according to the present first example is what is called a multifunction peripheral (MFP) having a plurality of functions such as a copy function, a print function, an image scanner function, and a fax function. FIG. **1** is a view of an internal configuration of the image forming apparatus **10** installed in a usable state, viewed from a front side of the image forming apparatus **10**. That is, an up-down direction in FIG. **1** corresponds to an up-down direction of the image forming apparatus **10**. A left-right direction in FIG. **1** corresponds to a left-right direction of the image forming apparatus **10**. Furthermore, a front side of the sheet surface in FIG. **1** corresponds to the front of the image forming apparatus **10**. A back side of the sheet surface in FIG. **1** corresponds to the rear of the image forming apparatus **10**.

[0032] An upper portion of the image forming apparatus **10** is provided with an image reader **12** as an image reader. The image reader **12** performs image read processing of reading an image of a document not illustrated and outputting two-dimensional read image data corresponding to the image of the document. Therefore, the image reader **12** includes a document table **14** on which a document is placed. The document table **14** is formed of a transparent member such as glass having a substantially rectangular flat plate shape, and is provided so that both main surfaces thereof are arranged along a horizontal direction. An image reading unit **16** is provided below the document table **14**. Although not described in detail, the image reading unit **16** includes a light source, a mirror, a lens, and a line sensor, and includes an image reading position Pr having a linear shape extending along a front-rear direction of the image forming apparatus **10** on an upper surface of the document table **14**. Furthermore, a drive mechanism not illustrated that causes the image reading position Pr of the image reading unit **16** to move (scan) along the left-right direction of the image forming apparatus **10** is provided below the document table **14**. That is, when the image reading position Pr of the image reading unit **16** is moved by the drive mechanism in a state where a document is placed on the document table **14**, an image of the document is read by what is called a fixed reading method. The front-rear direction of the image forming apparatus **10** is called a main scanning direction. The left-right direction of the image forming apparatus **10** is called a sub-scanning direction.

[0033] An automatic document feeder (ADF) **18** that also serves as a document pressing cover for pressing the document placed on the document table **14** is provided above the document table **14**. The automatic document feeder **18** is provided so as to be able to transition between a state of exposing the upper surface of the document table **14** to the outside and a state of covering the upper surface of the document table **14**. Therefore, the automatic document feeder **18** is joined to a main body (housing) of the image forming apparatus **10** via an appropriate movable support member such as a hinge not illustrated. FIG. **1** illustrates a state where the automatic document feeder **18** covers the upper surface of the document table **14**. The automatic document feeder **18** exhibits its original function when it is in a state of covering the upper surface of the document table **14** as

illustrated in FIG. 1.

[0034] The automatic document feeder **18** includes a document placement tray **20**. On the document placement tray **20**, a document, more precisely, a sheet-like document can be placed, and in particular, a plurality of documents can be placed in a stacked manner. Although not described in detail, the automatic document feeder **18** takes in documents placed on the document placement tray **20** one by one (one document at a time), and conveys the documents on a document conveyance path **22** in the automatic document feeder **18**. In the middle thereof, the document passes through the image reading position Pr, and more precisely, passes through the image reading position Pr that is in a fixed state. Thus, the image of the document is read by what is called a flow reading method. Afterwards, the document is discharged to a document discharge tray **24**.

[0035] An image former **26** as an image former is provided below the image reader **12**. The image former **26** performs image formation processing of forming an image based on appropriate image data such as the above-described read image data onto a sheet-like image recording medium not illustrated, for example, a paper sheet. That is, the image former **26** performs printing. This printing is performed by a known electrophotographic method. The image former **26** employs a tandem method to perform color printing.

[0036] Specifically, the image former **26** includes four process units (also called “image forming stations”) **28, 28, . . .** as monochrome toner image formers that individually form monochrome toner images, not illustrated, of a plurality of different colors, for example, four colors of yellow, magenta, cyan, and black. In addition, the image former **26** includes an exposure apparatus **30** as an exposure means that performs exposure necessary for formation of a monochrome toner image by each of the process units **28, 28, . . .**. Furthermore, the image former **26** includes a transfer unit **34** as a transfer means. The transfer unit **34** includes an intermediate transfer belt **32** (also called a “primary transfer belt”) to which the monochrome toner images formed by the respective process units **28, 28, . . .** are sequentially transferred, and transfers, onto a sheet, the toner image transferred to the intermediate transfer belt **32**. In addition, the image former **26** includes a fixing apparatus **36** as a fixer that fixes, on the sheet, the toner image transferred onto the sheet. Moreover, the image former **26** includes a toner replenishment apparatus **37** that replenishes toner not illustrated onto development apparatuses **50, 50, . . .** described later of the respective process units **28, 28, . . .**

[0037] First, the transfer unit **34** will be described in detail. The transfer unit **34** includes, in addition to the intermediate transfer belt **32** described above, a driving roller **38** that rotates the intermediate transfer belt **32**, and a driven roller **40** that stretches the intermediate transfer belt **32** together with the driving roller **38**. Furthermore, the transfer unit **34** includes four intermediate transfer rollers (also called “primary transfer rollers”) **42, 42, . . .** provided at positions corresponding to the respective process units **28, 28, . . .** on an inner side of the intermediate transfer belt **32**, and a transfer roller (also called “secondary transfer roller”) **44** as a transfer member.

[0038] The intermediate transfer belt **32** is stretched by the driving roller **38** and the driven roller **40**. The driving roller **38** rotates by receiving a driving force from a motor as an intermediate transfer belt driver not illustrated, and rotates counterclockwise in FIG. 1, for example. Together with this, the intermediate transfer belt **32** rotates (circulates) in the same direction, and the driven roller **40** also rotates in the same direction. A region **32a** on a lower side of the region between the driving roller **38** and the driven roller **40** in the intermediate transfer belt **32** is stretched along the horizontal direction, and the process units **28, 28, . . .** are arranged to face the region **32a** stretched along this horizontal direction. The region **32a** where the process units **28, 28, . . .** are arranged in the intermediate transfer belt **32** is called an intermediate transfer region. In this intermediate transfer region **32a**, the intermediate transfer belt **32** moves from the left side to the right side of the image forming apparatus **10**, that is, along the sub-scanning direction.

[0039] The intermediate transfer belt **32** is an endless belt-type body having flexibility and is made of a synthetic resin (e.g., polyimide or polycarbonate) to which a conductive material such as

carbon black is appropriately mixed. Although not described in detail, the driven roller **40** also has a slack prevention function of preventing slack of the intermediate transfer belt **32** by applying an appropriate tension to the intermediate transfer belt **32**.

[0040] The process units **28, 28, . . .** are provided at constant intervals below the intermediate transfer region **32a** of the intermediate transfer belt **32** along the movement direction of the intermediate transfer belt **32** in the intermediate transfer region **32a**, that is, along the sub-scanning direction. As described above, the process units **28, 28, . . .** each form, onto the intermediate transfer belt **32**, an individual monochrome toner image of four colors of yellow, magenta, cyan, and black. Although not apparent from the drawings including FIG. **1**, the process units **28, 28, . . .** are provided in order of yellow, magenta, cyan, and black from the upstream side to the downstream side (from the left side to the right side in FIG. **1**) in the movement direction of the intermediate transfer belt **32** in the intermediate transfer region **32a**. However, this order is merely an example while not limited thereto. The process units **28, 28, . . .** have the same structure as each other, except for each forming a monochrome toner image of a different color on the intermediate transfer belt **32**.

[0041] Each of the process units **28** includes a photoconductive drum **46**, a charging apparatus **48**, a development apparatus **50**, a cleaning apparatus **52**, and a static elimination apparatus not illustrated.

[0042] The photoconductive drum **46** is an image carrier that carries an electrostatic latent image and a monochrome toner image described later, and includes a cylindrical base formed of a conductive material such as aluminum. Although not illustrated, the base is grounded, and a photosensitive layer is formed on a surface (outer peripheral surface) of the base. The photoconductive drum **46** is provided in the intermediate transfer region **32a** so as to bring the surface of the base into contact with the outer surface of the intermediate transfer belt **32**, and rotates by receiving a driving force from a motor as a drum driver not illustrated. The rotation direction is clockwise in FIG. **1**. The photoconductive drum **46** rotates at a speed corresponding to the moving speed of the intermediate transfer belt **32**.

[0043] The charging apparatus **48** is a charger that charges the surface of the photoconductive drum **46** to a predetermined potential. The surface of the photoconductive drum **46** charged to a predetermined potential by this charging apparatus **48** is exposed by the exposure apparatus **30** described above, and the exposed region of the surface of the photoconductive drum **46** is lowered to a potential close to the ground potential (OV). The exposure apparatus **30** is provided below the line of the process units **28, 28, . . .** and exposes the surface of the photoconductive drum **46** of each of the process units **28** from below. That is, the exposure apparatus **30** irradiates the surface of the photoconductive drum **46** with light of an aspect in accordance with image data to be printed. Thus, an electrostatic latent image having a surface potential changed in accordance with image data to be printed is formed on the surface of the photoconductive drum **46**. The exposure apparatus **30** is a laser scanning unit including a laser diode not illustrated as a light source, for example, and a polygon mirror as a deflector. However, in place of this, an LED unit including an LED array in which LEDs as light sources are arranged may be employed as the exposure apparatus **30**.

[0044] The development apparatus **50** is a development unit that develops the electrostatic latent image formed on the surface of the photoconductive drum **46**. That is, the development apparatus **50** includes a developer accommodation chamber **50a**, and this developer accommodation chamber **50a** accommodates a developer containing toner. The developer accommodated in this developer accommodation chamber **50a** is stirred by stirring members **50c** and **50d** provided in the developer accommodation chamber **50a**. Thus, the toner contained in the developer is frictionally charged, and the development apparatus **50** develops the electrostatic latent image on the photoconductive drum **46** with this charged toner. In other words, the electrostatic latent image formed on the photoconductive drum **46** is visualized as a monochrome toner image by the toner contained in the

developer accommodated in the development apparatus **50** (developer accommodation chamber **50a**). The developer is a two-component developer containing a carrier in addition to the toner. The toner is a non-magnetic material, and the carrier is a magnetic material. A toner density sensor **50b** as a toner density detector a density (mass ratio of toner to carrier) T/D of toner contained in the developer in the developer accommodation chamber **50a** is provided below the developer accommodation chamber **50a**. This toner density sensor **50b** is what is called a permeability sensor, and detects the bulk density of the carrier contained in the developer by measuring the permeability of the developer, and eventually detects the toner density T/D.

[0045] The monochrome toner image visualized by development by this development apparatus **50** is transferred from the surface of the photoconductive drum **46** to the outer surface of the intermediate transfer belt **32** at a contact position between the surface of the photoconductive drum **46** and the outer surface of the intermediate transfer belt **32**, that is, the monochrome toner image is transferred by so-called intermediate transfer (primary transfer). Therefore, an intermediate transfer roller **42** is provided so as to face the photoconductive drums **46** across the intermediate transfer belt **32**. The intermediate transfer roller **42** is provided such that the surface (outer peripheral surface) thereof is in contact with the inner surface of the intermediate transfer belt **32**, and rotates by receiving a driving force generated by the rotation of the intermediate transfer belt **32**, that is, rotates counterclockwise in FIG. **1**. When the intermediate transfer roller **42** is applied with a predetermined intermediate transfer voltage from an intermediate transfer power source not illustrated, a transfer electric field is formed between the surface of the photoconductive drum **46** and the outer surface of the intermediate transfer belt **32**. By the action of this transfer electric field, the monochrome toner image on the photoconductive drum **46** is transferred onto the intermediate transfer belt **32**.

[0046] In this manner, monochrome toner images of the four colors of yellow, magenta, cyan, and black are individually formed on the intermediate transfer belt **32**. A color toner image is formed on the intermediate transfer belt **32** by the monochrome toner images of the four colors being superimposed on each other.

[0047] The (color) toner image formed on the intermediate transfer belt **32** is transferred to a sheet at a transfer nip Nt which is a contact part between the intermediate transfer belt **32** and the transfer roller **44**. Specifically, the transfer roller **44** is provided so as to press the intermediate transfer belt **32** between the transfer roller **44** and the driving roller **38** at a position facing the driving roller **38** across the intermediate transfer belt **32**. The transfer roller **44** rotates by receiving a driving force generated by the rotation of the intermediate transfer belt **32**, that is, rotates clockwise in FIG. **1**. Then, a transfer bias current having the same polarity as the toner charging polarity is applied from a transfer bias power source not illustrated to the driving roller **38**. Thus, a transfer electric field is formed between the intermediate transfer belt **32** and the transfer roller **44**, that is, at the transfer nip Nt. When the sheet passes through the transfer nip Nt in this state, the toner image on the intermediate transfer belt **32** is transferred onto the sheet.

[0048] The cleaning apparatus **52** is a cleaner that removes residual toner on the photoconductive drum **46** after the monochrome toner image is transferred from the photoconductive drum **46** onto the intermediate transfer belt **32**. The static elimination apparatus not illustrated is a static eliminator that removes static electricity on the photoconductive drum **46** after the residual toner is removed by the cleaning apparatus **52**. After the static electricity is removed by this static eliminator, the process subsequent to the charging by the charging apparatus **48** described above is repeated.

[0049] The fixing apparatus **36** is provided on a downstream side relative to the transfer nip Nt in a conveyance direction of the sheet conveyed along a sheet conveyance path **54** described later. As described above, the fixing apparatus **36** fixes, on the sheet, the toner image on the sheet, specifically, heats and melts the toner image, and further presses the toner image, thereby fixing (thermally fixing) the toner image on the sheet. Therefore, the fixing apparatus **36** includes a

heating belt **56**, a heating roller **58**, a fixing roller **60**, and a pressure roller **62**.

[0050] The heating belt **56** is an endless belt-type body having flexibility and is made of a synthetic resin (e.g., polyimide or polycarbonate) having high thermal conductivity. This heating belt **56** is stretched by the heating roller **58** and the fixing roller **60**. The heating roller **58** includes a cylindrical base (heat conductive layer) having high thermal conductivity, and a heat source is provided inside thereof. The heat source is a lamp heater such as a halogen lamp, and is heated by receiving supply of heating power from a heater heating power source not illustrated. The fixing roller **60** is a columnar roller member and includes a metal core and an elastic layer covering the metal core. That is, the heating roller **58** and the fixing roller **60** extend in a state of being parallel with each other, and are in contact within the heating belt **56**. The fixing roller **60** rotates by receiving a driving force from a motor as a heating belt driver not illustrated, and rotates counterclockwise in FIG. **1**, for example. Together with this, the heating belt **56** rotates (circulates) in the same direction as the fixing roller **60**, and the heating roller **58** also rotates in the same direction as the fixing roller **60**.

[0051] The pressure roller **62** is a columnar roller member and includes a metal core, an elastic layer covering the metal core, and a release layer covering the elastic layer. The pressure roller **62** is provided so as to press the heating belt **56** between the pressure roller **62** and the fixing roller **60** at a position facing the fixing roller **60** across the heating belt **56**. That is, the pressure roller **62** is provided so as to extend along a rotation axis direction of the fixing roller **60**, in other words, so as to extend along a rotation axis direction of the heating belt **56**. This pressure roller **62** rotates by receiving a driving force generated by the rotation of the heating belt **56**, that is, rotates clockwise in FIG. **1**. A fixing nip Nf which is a pressing part of the heating belt **56** by the pressure roller **62** is positioned in the sheet conveyance path **54** described later. When the sheet passes through this fixing nip Nf, the toner image on the sheet is heated and melted, further pressed, and fixed on the sheet.

[0052] Although not described in detail including drawings, the fixing apparatus **36** includes a temperature sensor that detects a surface temperature (fixing temperature) of the heating belt **56**. Based on a detection result by this temperature sensor, the heating temperature of the heating roller **58** by the above-described heat source is controlled, and thus, the surface temperature of the heating belt **56** is controlled. For example, a thermistor is used as the temperature sensor, but the temperature sensor is not limited to the thermistor. In the fixing apparatus **36**, a configuration may be employed in which the heating belt **56** is not provided, the fixing roller **60** also serves as the heating roller **58**, and a direct contact part between the fixing roller **60** and the pressure roller **62** serves as the fixing nip Nf.

[0053] The toner replenishment apparatus **37** replenishes toner to each of the development apparatuses **50**, **50**, . . . , more specifically, to the developer accommodation chamber **50a** of each of the development apparatuses **50**. That is, when development is performed by each development apparatus **50**, the toner in the developer accommodation chamber **50a** of each development apparatus **50** is consumed. In order to compensate for consumption of the toner, the toner replenishment apparatus **37** replenishes the toner to the developer accommodation chamber **50a** of each development apparatus **50**. Although the toner replenishment apparatus **37** will be described in detail later, the toner replenishment apparatus **37** is mounted with four toner cartridges **100**, **100**, . . . accommodating toner corresponding to the above-described four colors.

[0054] Furthermore, a sheet feeder **64** as a sheet feed means is provided below the image former **26**, in other words, below the image forming apparatus **10**. The sheet feeder **64** includes a sheet feeding cassette **66**, and a plurality of sheets can be accommodated in this sheet feeding cassette **66** in a stacked manner. In addition, the sheet feeder **64** includes a pickup roller **68**. The sheet feeder **64** takes out, one by one by the pickup roller **68**, sheets accommodated in the sheet feeding cassette **66**, and supplies the sheets to the sheet conveyance path **54** described next.

[0055] The sheet conveyance path **54** is provided from the sheet feeder **64**, via the transfer nip Nt

and the fixing nip Nf, to a sheet discharge port **72** leading to a sheet discharge tray **70**. Appropriate positions of the sheet conveyance path **54** are provided with a plurality of conveyance rollers (strictly speaking, roller pairs) **74, 74, . . .** that convey sheets along the sheet conveyance path **54** from the sheet feeder **64** toward the sheet discharge port **72**. Among the conveyance rollers **74, 74, . . .**, a conveyance roller **74a** provided at a position closest to the transfer nip Nt on an upstream side relative to the transfer nip Nt in the conveyance direction of the sheet in the sheet conveyance path **54** is a resist roller (also called “paper stop roller”) that measures a timing at which the sheet passes through the transfer nip Nt. Of the conveyance rollers **74, 74, . . .**, a conveyance roller **74b** provided at the most distal downstream position in the conveyance direction of the sheet in the sheet conveyance path **54**, that is, in the vicinity of the sheet discharge port **72**, is a sheet discharge roller that discharges the sheet to the sheet discharge tray **70** via the sheet discharge port **72**. The sheet discharge tray **70** is provided between the image reader **12** and the image former **26**, that is, provided in what is called an in-body space while not limited thereto.

[0056] In addition, a reverse conveyance path **76** for double-sided printing is provided in the image forming apparatus **10**. This reverse conveyance path **76** is a conveyance path in which a sheet that has once passed through the fixing nip Nf, that is, a sheet after printing is taken in, and the sheet is provided for printing again. That is, the sheet taken into the reverse conveyance path **76** is supplied to the sheet conveyance path **54** again via the reverse conveyance path **76**, and specifically, is supplied to the upstream side of a resist roller **74a**. Thus, the sheet supplied to the upstream side of the resist roller **74a** is brought into a state where the front and back of the sheet are reversed. Printing is performed on this sheet that has been turned over, and what is called double-sided printing is achieved. An appropriate position of the reverse conveyance path **76** is also provided with a conveyance roller **78**.

[0057] A right side surface of the image forming apparatus **10** is provided with a manual feed tray **80**. A plurality of sheets can be placed on this manual feed tray **80** in a stacked manner. When this manual feed tray **80** is designated as a sheet supply source, the sheet feeder **64** supplies sheets one by one from the manual feed tray **80** to the sheet conveyance path **54**.

[0058] Moreover, the sheet feeder **64** may include an optional sheet feeding cassette not illustrated. This optional sheet feeding cassette is provided below the sheet feeding cassette **66**. When the optional sheet feeding cassette is designated as a sheet supply source, the sheet feeder **64** supplies sheets one by one from the optional sheet feeding cassette to the sheet conveyance path **54**. An optional conveyance roller **82** that supplies a sheet from the optional sheet feeding cassette to the sheet conveyance path **54** is provided at an appropriate position.

[0059] Next, the toner replenishment apparatus **37** will be described in detail with reference to FIGS. **2** to **4**. Here, FIG. **2** is a perspective view of the toner replenishment apparatus **37** as viewed from the front right oblique upper side. FIG. **3** is a cross-sectional view of the toner replenishment apparatus as viewed from an obliquely upper right front side at an angle slightly different from that in FIG. **2**. FIG. **4** is a view illustrating a transmission path of a rotational force from a drive source **110** described later.

[0060] The toner replenishment apparatus **37** included in the image forming apparatus **10** according to the first example includes two replenishment apparatuses of a first replenishment apparatus **37a** and a second replenishment apparatus **37b**. Of them, the first replenishment apparatus **37a** is responsible for toner replenishment of two colors of yellow and magenta. For this purpose, the first replenishment apparatus **37a** includes the toner cartridge **100** (hereinafter, may be called “yellow toner cartridge **100[Y]**”) as an accommodation for yellow toner in which yellow toner is accommodated, the toner cartridge **100** (hereinafter, may be called “magenta toner cartridge **100[M]**”) as an accommodation for magenta toner in which magenta toner is accommodated, a first replenishment motor **110a** that is the drive source **110** that replenishes toner from the toner cartridges **100[Y]** and **100[M]** to the respective development apparatuses **50**, and internal replenishment paths **104 [Y]** and **104 [M]** through which toner is guided from the toner cartridges

100[Y] and **100[M]** to the respective development apparatuses **50**. The toner cartridges **100[Y]** and **100[M]**, the first replenishment motor **110a**, and the internal replenishment paths **104 [Y]** and **104 [M]** are attached to the main body of the image forming apparatus **10**, but the toner cartridges **100[Y]** and **100[M]** are easily and detachably attached to the main body of the image forming apparatus **10** so as to be replaceable when the toner therein runs out.

[0061] On the other hand, the second replenishment apparatus **37b** is responsible for toner replenishment of two colors of cyan and black. For this purpose, the second replenishment apparatus **37b** includes the toner cartridge **100** (hereinafter, may be called “cyan toner cartridge **100[C]**”) as an accommodation for cyan toner in which cyan toner is accommodated, the toner cartridge **100** (hereinafter, may be called “black toner cartridge **100[K]**”) as an accommodation for black toner in which black toner is accommodated, a second replenishment motor **110b** that is the drive source **110** that replenishes toner from the toner cartridges **100[C]** and **100[K]** to the respective development apparatuses **50**, and internal replenishment paths **104 [C]** and **104 [K]** through which toner is guided from the toner cartridges **100[C]** and **100[K]** to the respective development apparatuses **50**. The toner cartridges **100[C]** and **100[K]**, the second replenishment motor **110b**, and the internal replenishment paths **104 [C]** and **104 [K]** are attached to the main body of the image forming apparatus **10**, but the toner cartridges **100[C]** and **100[K]** are easily and detachably attached to the main body of the image forming apparatus **10** so as to be replaceable when the toner therein runs out.

[0062] Each of the toner cartridges **100** is internally provided with a conveyance replenishment member (auger screw) **102** that conveys the toner accommodated in the toner cartridge **100** forward while stirring the toner (FIG. 3). A first gear **116** is attached to a front end portion of each of the conveyance replenishment members **102** (FIG. 2), and when a rotational force from the corresponding drive source **110** is transmitted to the first gear **116**, the conveyance replenishment member **102** rotates. When this conveyance replenishment member **102** rotates, the toner accommodated in the toner cartridge **100** is conveyed forward and sent to the internal replenishment path **104** via a discharge port **118** provided in front of the toner cartridge **100**. The internal replenishment path **104** provided so as to correspond to the discharge port **118** of each toner cartridge **100** is provided with a replenishment member (screw feeder) **106**. Each replenishment member **106** is also provided with a second gear **120** that receives the rotational force from the corresponding drive source **110** at an end (end in the left direction), and the second gear **120** receives the rotational force from the drive source **110** and rotates at a constant speed, whereby the toner supplied from each toner cartridge **100** via each discharge port **118** is discharged from a replenishment port **108** at a constant speed ($V_s[X]$). The toner discharged from this replenishment port **108** is replenished to the developer accommodation chamber **50a** of the corresponding development apparatus **50** via an external replenishment path not illustrated. Since the black toner is greater in consumption amount than the toner of other colors, the black toner cartridge **100[K]** may be larger in size than each of the other toner cartridges **100**.

[0063] In the first example, the replenishment member **106** is provided between the internal replenishment path **104** and the replenishment port **108**, but the internal replenishment path **104** may be configured such that the replenishment member **106** is eliminated and the discharge port **118** is positioned directly above the replenishment port **108**. In this case, the toner in the toner cartridge **100** is replenished to the developer accommodation chamber **50a** by the rotation of the conveyance replenishment member **102**.

[0064] Here, the first replenishment apparatus **37a** in the first example selectively drives, by the first replenishment motor **110a** which is a common (one) drive source, any one of the replenishment member **106** (yellow replenishment member **106[Y]**) for the yellow toner cartridge **100[Y]** and the conveyance replenishment member **102** (yellow conveyance replenishment member **102[Y]**) in the yellow toner cartridge **100[Y]**, and the replenishment member **106** (magenta replenishment member **106[M]**) for the magenta toner cartridge **100[M]** and the conveyance

replenishment member **102** (magenta conveyance replenishment member **102[M]**) in the magenta toner cartridge **100[M]**. That is, the first replenishment motor **110a** selectively drives the yellow replenishment member **106[Y]** and the yellow conveyance replenishment member **102[Y]**, and the magenta replenishment member **106[M]** and the magenta conveyance replenishment member **102[M]**. In other words, the first replenishment motor **110a** cannot simultaneously drive the yellow replenishment member **106[Y]** and the yellow conveyance replenishment member **102[Y]**, and the magenta replenishment member **106[M]** and the magenta conveyance replenishment member **102[M]**.

[0065] More specifically, with reference to FIG. 4, a rotation shaft **110a** of the first replenishment motor **111a** is provided with a drive gear **112a** (in FIG. 4, the first replenishment motor **110a** is not illustrated so that the arrangement of the drive gear **112a** can be seen). The drive gear **112a** is connected with a yellow gear train **121[Y]** that transmits a rotational force to the first gear **116** (yellow first gear **116[Y]**) attached to the yellow conveyance replenishment member **102[Y]** and the second gear **120** (yellow second gear **120[Y]**) attached to the yellow replenishment member **106[Y]**, and a magenta gear train **121[M]** that transmits a rotational force to the first gear **116** (magenta first gear **116[M]**) attached to the magenta conveyance replenishment member **102[M]** and the second gear **120** (magenta second gear **120[M]**) attached to the magenta replenishment member **106[M]**. Each of the yellow gear train **121[Y]** and the magenta gear train **121[M]** is provided with one one-way clutch not illustrated in which the direction where the rotational force is transmitted is one direction. Here, the one-way clutch provided in the yellow gear train **121[Y]** is arranged so as to transmit the rotational force to the yellow gear train **121[Y]** when the drive gear **112a** of the first replenishment motor **110a** rotates in a predetermined direction R1 (counterclockwise in FIG. 4), but not to transmit the rotational force to the yellow gear train **121[Y]** when the drive gear **112a** rotates in a direction R2 (clockwise in FIG. 4) opposite to the predetermined direction R1. Similarly, the one-way clutch provided in the magenta gear train **121[M]** is arranged so as not to transmit the rotational force to the magenta gear train **121[M]** when the drive gear **112a** rotates in one direction R1 which is a predetermined direction, and so as to transmit the rotational force to the magenta gear train **121[M]** when the drive gear **112a** rotates in the other direction R2 which is a direction opposite to the one direction R1. That is, when (the drive gear **112a** of) the first replenishment motor **110a** rotates in the one direction R1, the yellow replenishment member **106[Y]** and the yellow conveyance replenishment member **102[Y]** are driven, whereby the yellow toner is replenished to the developer accommodation chamber **50a** of the development apparatus **50** for yellow. When the first replenishment motor **110a** rotates in the other direction R2, the magenta replenishment member **106[M]** and the magenta conveyance replenishment member **102[M]** are driven, whereby the magenta toner is replenished to the developer accommodation chamber **50a** of the development apparatus **50** for magenta. FIG. 4 is a view illustrating an extracted main part of the first replenishment apparatus **37a**. The first replenishment motor **110a** is, for example, a stepping motor, but is not limited to this.

[0066] Similarly, the second replenishment apparatus **37b** in the first example also selectively drives, by the second replenishment motor **110b** which is a common (one) drive source, any one of the replenishment member **106** (cyan replenishment member **106[C]**) for the cyan toner cartridge **100[C]** and the conveyance replenishment member **102** (cyan conveyance replenishment member **102[C]**) in the cyan toner cartridge **100[C]**, and the replenishment member **106** (black replenishment member **106[K]**) for the black toner cartridge **100[K]** and the conveyance replenishment member **102** (black conveyance replenishment member **102[K]**) in the black toner cartridge **100[K]**. That is, the second replenishment motor **110b** selectively drives the cyan replenishment member **106[C]** and the cyan conveyance replenishment member **102[C]**, and the black replenishment member **106[K]** and the black conveyance replenishment member **102[K]**. In other words, the second replenishment motor **110b** cannot simultaneously drive the cyan replenishment member **106[C]** and the cyan conveyance replenishment member **102[C]**, and the

black replenishment member **106**[K] and the black conveyance replenishment member **102**[K]. [0067] A rotation shaft **111b** of the second replenishment motor **110b** is also provided with a drive gear **112b** similar to that provided on the rotation shaft **111a** of the first replenishment motor **110a** (in FIG. 4, the rotation shaft **111b** and the drive gear **112b** of the second replenishment motor **110b** are blocked by the second replenishment motor **110b** and do not appear). The drive gear **112b** of the second replenishment motor **110b** is also connected with a cyan gear train that constitutes a transmission path through which a rotational force is transmitted to the first gear **116** (cyan first gear **116**[C]) attached to the cyan conveyance replenishment member **102**[C] and the second gear **120** (cyan second gear **120**[C]) attached to the cyan replenishment member **106**[C] and is provided with a one-way clutch in the transmission path, and a black gear train that constitutes a transmission path through which a rotational force is transmitted to the first gear **116** (black first gear **116**[K]) attached to the black conveyance replenishment member **102**[K] and the second gear (black second gear) attached to the black replenishment member **106**[K] and is provided with a one-way clutch in the transmission path. Similarly to the first replenishment motor **110a**, when (the drive gear **112b** of) the second replenishment motor **110b** rotates in the one direction R1, the cyan replenishment member **106**[C] and the cyan conveyance replenishment member **102**[C] are driven, and when the drive gear **112b** rotates in the other direction R2, the black replenishment member **106**[K] and the black conveyance replenishment member **102**[K] are driven to perform replenishment of the respective toner. Similarly to the first replenishment motor **110a**, the second replenishment motor **110b** is also a stepping motor, for example, but is not limited thereto.

[0068] As described above, when each of the replenishment members **106** rotates at a constant speed, the developer accommodation chamber **50a** of the corresponding development apparatus **50** is replenished with toner at a constant speed ($V_s[X]$), but a replenishment speed $V_s[X]$ (X ; index representing any of Y, M, C, and K) of this toner has a value at which the toner is not replenished in a lump, and a necessary and sufficient stirring time for charging the replenished toner as intended can be secured. The replenishment speed $V_s[X]$ of the toner of the toner replenishment apparatus **37** in the first example is expressed as a toner amount that can be replenished in a unit time, and the value thereof is, for example, 0.2 g/s. This value can be obtained by experiment, and the replenishment apparatus **37** in the first example drives the replenishment member **106** (and the conveyance replenishment member **102**) so that 0.2 g of toner is replenished per second from the replenishment port **108** (and the discharge port **118**). The amount of toner to be replenished by the toner replenishment apparatus **37** during toner image formation can be controlled (adjusted) by the drive time of the drive source **110** that drives the replenishment member **106** (and the conveyance replenishment member **102**). Next, a toner consumption amount $Q_c[X]$ consumed along with the formation of the toner image will be described.

[0069] The toner consumption amount $Q_c[X]$ can be calculated by the size of the sheet on which the toner image is formed and a printing rate $R_p[X]$. Here, the toner consumption amount $Q_c[X]$ that is actually consumed when the size of the sheet on which the toner image is formed is the A4 size and a toner image in which the printing rate $R_p[X]$ which is the area ratio of the toner image to the size of the sheet is 5% is formed is obtained from the experimental result, and the value thereof is 0.013 g per sheet. Since the toner consumption amount $Q_c[X]$ has a correlation with the printing rate $R_p[X]$, the toner consumption amount $Q_c[X]$ when the size of the sheet is the A4 size and the printing rate $R_p[X]$ is 100% can be predicted to be 0.26 g per sheet by proportional calculation since the consumption amount $Q_c[X]$ at the printing rate $R_p[X]$ 5% is 0.013 g. Thus, since the toner consumption amount $Q_c[X]$ can be predicted by the value of the printing rate $R_p[X]$, it is understood that the toner replenishment amount to be replenished by the toner replenishment apparatus **37** may be performed based on the printing rate $R_p[X]$. The printing rate $R_p[X]$ is obtained by being converted into the A4 size. Therefore, for example, in a case of the A3 size with the printing rate $R_p[X]$ 100%, since the area of A3 is twice of A4, conversion is performed to 200%, that is, for two sheets of A4 size, but thereafter, the size of the sheet is assumed to be the A4

size unless otherwise specified. This toner consumption amount $Qc[X]$ is also common (same) in all the four colors.

[0070] Next, the number of printed sheets per minute (CPM) representing the printing speed of the image forming apparatus **10** in the first example is, for example, 35. Therefore, a one-sheet printing time (sheet passing time) Tp ($=1/CPM$), which is the time required to print one sheet of paper, is about 1.71 s. That is, when an A4 size toner image with the printing rate $Rp[X]$ of 100% is formed, 0.26 g of toner needs to be replenished, but since the toner replenishment rate $Vs[X]$ per unit time of the replenishment apparatus **37** is 0.2 g/S, the time (drive time) required to replenish 0.26 g of toner is 1.3 seconds, and the toner replenishment can be finished within the one-sheet printing time.

[0071] Here, focusing on the first replenishment apparatus **37a**, according to the first replenishment apparatus **37a**, the yellow replenishment member **106[Y]** and the magenta replenishment member **106[M]** are selectively driven by the first replenishment motor **110a** as described above, that is, the replenishment of the yellow toner and the replenishment of the magenta toner are selectively performed. Therefore, for example, when the printing rate $Rp[Y]$ of yellow related to printing on one sheet of paper is 5% and the printing rate $Rp[M]$ of magenta is 5%, that is, when the consumption amount $Qc[Y]$ of the yellow toner is 0.013 g and the consumption amount $Qc[M]$ of the magenta toner is also 0.013 g, the drive time of the yellow replenishment member **106[Y]** necessary for compensating for the consumption of the yellow toner, that is, the necessary replenishment time $Ts[Y]$ ($=Qc[Y]/Vs[Y]$) of the yellow toner is 0.065 s, and the necessary replenishment time $Ts[M]$ ($=Qc[M]/Vs[M]$) of the magenta toner, which is the drive time of the magenta replenishment member **106[M]** for compensating for the consumption of the magenta toner, is also 0.065 s. That is, in order to compensate for the consumption of the yellow toner and the magenta toner by printing, a total time of 0.13 s is required. This time of 0.13 s is shorter than the printing time Tp per sheet of about 1.71 s described above. This means that it is possible to replenish all consumption of the yellow toner and the magenta toner by the printing within the printing time Tp per sheet, in other words, while printing is performed on one sheet.

[0072] On the other hand, for example, when the printing rate $Rp[Y]$ of yellow related to printing on one sheet is 100% (yellow solid) and the printing rate $Rp[M]$ of magenta is 100% (magenta solid), that is, when the consumption amount $Qc[Y]$ of the yellow toner is 0.26 g and the consumption amount $Qc[M]$ of the magenta toner is also 0.26 g, the necessary replenishment time $Ts[Y]$ for compensating for the consumption of the yellow toner is 1.3 s, and the necessary replenishment time $Ts[M]$ for compensating for the consumption of the magenta toner is also 1.3 s. That is, in order to compensate for the consumption of the yellow toner and the magenta toner by printing, a total time of 2.6 s is required. This time of 2.6 s is longer than the printing time Tp per sheet of about 1.71 s. This means that it is not possible to replenish all consumption of the yellow toner and the magenta toner by the printing within the printing time Tp per sheet, that is, while printing is performed on one sheet, so to speak, the replenishment is not in time.

[0073] Therefore, it is important to drive the yellow replenishment member **106[Y]** and the magenta replenishment member **106[M]** in an appropriate balance according to the necessary replenishment time $Ts[Y]$ of the yellow toner and the necessary replenishment time $Ts[M]$ of the magenta toner, strictly, according to an accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and an accumulation necessary replenishment time $Tz[M]$ of the magenta toner. The accumulation necessary replenishment time $Tz[Y]$ of the yellow toner is an accumulation value of the shortage in which the drive time (replenishment time of yellow toner) of the actual yellow replenishment member **106[Y]** is subtracted from the necessary replenishment time $Ts[Y]$ of the yellow toner. The accumulation necessary replenishment time $Tz[M]$ of the magenta toner is an accumulation value of a shortage of the drive time (replenishment time of the magenta toner) in which the actual magenta replenishment member **106[M]** is subtracted from the necessary replenishment time $Ts[M]$ of the magenta toner. These accumulation necessary replenishment times $Tz[Y]$ and $Tz[M]$ are obtained each time printing is performed.

[0074] Therefore, in the present first example, for example, when the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner becomes equal to or greater than a toner replenishment threshold Ta as a predetermined first threshold, the yellow replenishment member **106**[Y] is driven, that is, replenishment of the yellow toner is performed. This replenishment of the yellow toner is performed for the accumulation necessary replenishment time $Tz[Y]$ with a continuous replenishment allowable time Ton as a predetermined second threshold as an upper limit. That is, when the accumulation necessary replenishment time $Tz[Y]$ is equal to or less than the continuous replenishment allowable time Ton , replenishment of the yellow toner is performed for the accumulation necessary replenishment time $Tz[Y]$. On the other hand, when the accumulation necessary replenishment time $Tz[Y]$ exceeds the continuous replenishment allowable time Ton , the replenishment of the yellow toner is performed for the continuous replenishment allowable time Ton and then the replenishment of the yellow toner is temporarily stopped. In other words, the replenishment of the yellow toner is temporarily stopped before the replenishment of the yellow toner for the accumulation necessary replenishment time $Tz[Y]$ is completed. After a predetermined replenishment stop time $Toff$ has elapsed, the replenishment of the yellow toner is resumed on condition that the accumulation necessary replenishment time $Tz[M]$ of the magenta toner which is the other toner is less than the toner replenishment threshold Ta . This replenishment of the yellow toner is performed until the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner reaches 0 (zero), strictly, equal to or less than 0. In short, replenishment of the yellow toner is intermittently performed in a pattern in accordance with what is called a replenishment duty ($Ton: Toff$) formed by a combination of the continuous replenishment allowable time Ton and the replenishment stop time $Toff$.

[0075] Similarly to this, also when the accumulation necessary replenishment time $Tz[M]$ of the magenta toner becomes equal to or greater than the toner replenishment threshold Ta , the magenta replenishment member **106**[M] is driven, that is, replenishment of the magenta toner is performed. This replenishment of the magenta toner is performed for the accumulation necessary replenishment time $Tz[M]$ with the continuous replenishment allowable time Ton as an upper limit. In particular, when the accumulation necessary replenishment time $Tz[M]$ exceeds the continuous replenishment allowable time Ton , replenishment of the magenta toner is intermittently performed in a pattern in accordance with the above-described replenishment duty on condition that the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner, which is the other toner, is less than the toner replenishment threshold Ta .

[0076] When both the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner become equal to or greater than the toner replenishment threshold Ta , first, the toner replenishment related to the longer one of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner (i.e., the one having a high necessity of replenishment) is preferentially performed. That is, similarly to that described above, the replenishment of one toner α is performed for the accumulation necessary replenishment time $Tz[a]$ (a ; index representing Y or M) of the one toner α with the continuous replenishment allowable time Ton as an upper limit. In other words, the replenishment of the one toner α is performed for the accumulation necessary replenishment time $Tz[a]$ or the continuous replenishment allowable time Ton . Thereafter, replenishment of another toner β is performed. This replenishment of the other toner β is also performed for the accumulation necessary replenishment time $Tz[\beta]$ (β ; index representing M or Y) of the other toner β with the continuous replenishment allowable time Ton as an upper limit. Thereafter, according to the current accumulation necessary replenishment time $Tz[a]$ of the one toner α and the current accumulation necessary replenishment time $Tz[\beta]$ of the other toner β , the replenishment of the one toner α is resumed, the replenishment of the other toner β is resumed, or both the replenishment of the one toner α and the replenishment of the other toner β are brought into a stop state.

[0077] That is, according to the present first example, when at least one of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner becomes equal to or greater than the toner replenishment threshold Ta , replenishment of the one toner α related to the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner or the accumulation necessary replenishment time $Tz[M]$ of the magenta toner that becomes equal to or greater than the toner replenishment threshold Ta is performed. This replenishment of the one toner α is performed for the accumulation necessary replenishment time $Tz[a]$ of the one toner α with the continuous replenishment allowable time Ton as an upper limit. Thereafter, according to the accumulation necessary replenishment time $Tz[\beta]$ of the other toner β and the current accumulation necessary replenishment time $Tz[a]$ of the one toner α , the replenishment of the other toner β is performed, the replenishment of the one toner α is resumed, or the replenishment of both of these toners is brought into a stop state. In particular, when the replenishment of the other toner β is performed, the replenishment of the other toner β is performed for the accumulation necessary replenishment time $Tz[\beta]$ of the other toner β with the continuous replenishment allowable time Ton as the upper limit. When the replenishment of the one toner α is resumed, after the predetermined replenishment stop time $Toff$ has elapsed, the replenishment of the one toner α is resumed for the current accumulation necessary replenishment time $Tz[a]$ of the one toner α with the continuous replenishment allowable time Ton as an upper limit. When the accumulation necessary replenishment time $Tz[\beta]$ of the other toner β is equal to or less than the toner replenishment threshold Ta and the current accumulation necessary replenishment time $Tz[a]$ of the one toner α is also equal to or less than the toner replenishment threshold Ta , the replenishment of both of the toners α and β is brought into a stop state.

[0078] The toner replenishment threshold Ta has an appropriate value, for example, 1 s, so that toner replenishment is not frequently performed, in other words, the toner replenishment is performed at a necessary and sufficient frequency. The continuous replenishment allowable time Ton preferably has a value equal to or greater than the necessary replenishment time $Ts[X]$ when the printing rate $Rp[X]$ is 50% and equal to or less than the necessary replenishment time $Ts[X]$ when the printing rate $Rp[X]$ is 90%, more preferably has a value equal to or greater than the necessary replenishment time $Ts[X]$ when the printing rate $Rp[X]$ is 75% and equal to or less than the necessary replenishment time $Ts[X]$ when the printing rate $Rp[X]$ is 85%, for example, 1 s. The continuous replenishment allowable time Ton of 1 s corresponds to the necessary replenishment time $Ts[X]$ when the printing rate $Rp[X]$ is about 77%. The replenishment stop time $Toff$ has a value of about $1/20$ to $1/5$ of the continuous replenishment allowable time Ton , for example, 0.1 s corresponding to a value of $1/10$ of the continuous replenishment allowable time Ton . Specific values of the toner replenishment threshold Ta , the continuous replenishment allowable time Ton , and the replenishment stop time $Toff$ are merely examples, and are not limited to these values.

[0079] Furthermore, when both the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner become equal to or greater than the toner replenishment threshold Ta , and when the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner have the same length as each other, a toner density detection value $T/D[Y]$ by the toner density sensor **50b** of the development apparatus **50** for yellow and a toner density detection value $T/D[M]$ by the toner density sensor **50b** of the development apparatus **50** for magenta are compared with each other. The toner replenishment related to the smaller value (one with a lower toner density T/D) of the toner density detection value $T/D[Y]$ of the yellow toner and the toner density detection value $T/D[M]$ of the magenta toner is preferentially performed. Thereafter, the same applies to a case where both the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner become equal to or greater than the toner replenishment threshold Ta , and where first, the toner replenishment related to the longer one of the accumulation necessary

replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner is preferentially performed.

[0080] When both the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner become equal to or greater than the toner replenishment threshold Ta , and when the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner have the same length as each other, and the toner density detection value $T/D[Y]$ of the yellow toner and the toner density detection value $T/D[M]$ of the magenta toner have the same value as each other, the replenishment of the toner on a predetermined side, for example, the replenishment of the yellow toner is preferentially performed. Thereafter, the same applies to a case where both the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner become equal to or greater than the toner replenishment threshold Ta , and where first, the toner replenishment related to the longer one of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner is preferentially performed. Not the replenishment of the yellow toner but the replenishment of the magenta toner may be preferentially performed.

[0081] Here, for example, assume a case where printing in which the printing rate $Rp[Y]$ of yellow is 100% and the printing rate $Rp[M]$ of magenta is 100% is continuously performed. In this case, as illustrated in FIG. 5, printing on one sheet of paper is repeatedly performed at a cycle of the printing time Tp being 1.71 s. On the other hand, the replenishment of the yellow toner and the replenishment of the magenta toner are alternately performed at a cycle of the continuous replenishment allowable time Ton being 1 s, so to speak, performed totally asynchronously with the printing. As described above, the continuous replenishment allowable time Ton of 1 s corresponds to the necessary replenishment time $Ts[X]$ in a case where the printing rate $Rp[X]$ is about 77%, and the replenishment of the yellow toner and the replenishment of the magenta toner are alternately performed at a cycle of such the continuous replenishment allowable time Ton , whereby the replenishment of the yellow toner and the replenishment of the magenta toner are performed in an extremely good (perfect) balance. Thus, the balance between the toner density $T/D[Y]$ in the developer accommodation chamber 50a of the development apparatus 50 for yellow and the toner density $T/D[M]$ in the developer accommodation chamber 50a of the development apparatus 50 for magenta is well kept, and furthermore, a good density balance is also kept on an image (output image) finally formed on a sheet. FIG. 5 illustrates an example in which replenishment of the magenta toner is preferentially performed.

[0082] The replenishment manner of the yellow toner and the magenta toner illustrated in FIG. 5 will be described with reference to FIG. 6 in relation to the respective accumulation necessary replenishment times $Tz[X]$ ($Tz[Y]$ and $Tz[M]$). FIG. 6 illustrates an example in which the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner immediately before a job instructing execution of printing is received is 0 s, the accumulation necessary replenishment time $Tz[M]$ of the magenta toner is also 0 s, and the replenishment of the magenta toner is preferentially performed.

[0083] As illustrated in FIG. 6, when printing of the first sheet is performed, the accumulation necessary replenishment time $Tz[M]$ of the magenta toner is 1.3 s, and the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner is also 1.3 s. More specifically, the printing rate $Rp[M]$ of magenta and the printing rate $Rp[Y]$ of yellow are derived based on the image data (pixel data) used for printing of the first sheet. The necessary replenishment time $Ts[M]$ of the magenta toner is derived based on the printing rate $Rp[M]$ of magenta, and the necessary replenishment time $Ts[Y]$ of the yellow toner is derived based on the printing rate $Rp[Y]$ of yellow. The accumulation necessary replenishment time $Tz[M]$ of the magenta toner is updated by the necessary replenishment time $Ts[M]$ of the magenta toner that is derived, and the accumulation necessary

replenishment time $Tz[Y]$ of the yellow toner is updated by the necessary replenishment time $Ts[Y]$ of the yellow toner that is derived. As a result, the accumulation necessary replenishment time $Tz[M]$ of the magenta toner has a value of 1.3 s in which the necessary replenishment time $Ts[M]$ of the magenta toner of 1.3 s according to the printing rate $Rp[M]$ of magenta of 100% is added to the value of 0 s before the update. The accumulation necessary replenishment time $Tz[Y]$ of the yellow toner has a value of 1.3 s in which the necessary replenishment time $Ts[Y]$ of the yellow toner of 1.3 s corresponding to the printing rate $Rp[Y]$ of yellow of 100% is added to the value of 0 s before the update.

[0084] Since both the accumulation necessary replenishment time $Tz[M]$ ($=1.3$ s) of the magenta toner and the accumulation necessary replenishment time $Tz[Y]$ ($=1.3$ s) of the yellow toner updated in this manner are equal to or greater than the toner replenishment threshold Ta ($=1$ s), first, the replenishment of one of the magenta toner and the yellow toner is performed, and here, the replenishment of the magenta toner is preferentially performed as described above. This replenishment of the magenta toner is performed for the accumulation necessary replenishment time $Tz[M]$ of the magenta toner with the continuous replenishment allowable time Ton ($=1$ s) as an upper limit, that is, for the continuous replenishment allowable time Ton of 1 s. Accordingly, the accumulation necessary replenishment time $Tz[M]$ of the magenta toner becomes 0.3 s ($=1.3$ s -1 s). On the other hand, the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner, which is 1.3 s, remains unchanged.

[0085] After this replenishment of the magenta toner is performed, the replenishment of the yellow toner which is the other toner is performed. This replenishment of the yellow toner is performed for the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner with the continuous replenishment allowable time Ton as an upper limit, that is, for the continuous replenishment allowable time Ton of 1 s. On the other hand, while the replenishment of the yellow toner is being performed, the printing of the first sheet is finished, and the printing of the second sheet is performed. That is, the printing of the first sheet is performed until a time of 0.71 s ($=Tp-1$ s) elapses from the start of the replenishment of the yellow toner, and the printing of the second sheet is performed after the time of 0.71 s elapses. Therefore, the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner at the time point when the printing of the first sheet is finished is 0.59 s ($=1.30$ s -0.71 s). The accumulation necessary replenishment time $Tz[M]$ of the magenta toner at the time point when the printing of the first sheet is finished, which is 0.3 s, remains unchanged.

[0086] When printing of the second sheet is performed, the accumulation necessary replenishment time $Tz[M]$ of the magenta toner is updated to 1.6 s ($=0.3$ s $+1.3$ s), and the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner is updated to 1.89 s ($=0.59$ s $+1.3$ s). At this time, the replenishment of the yellow toner is in a state of being continuously performed. Then, at a time point when a time of 0.29 s ($=1$ s -0.71 s) has elapsed from the start of printing of the second sheet, the replenishment of the yellow toner for a time of 1 s is finished. The accumulation necessary replenishment time $Tz[Y]$ of the yellow toner at this time point is 1.6 s ($=1.89$ s -0.29 s), and the accumulation necessary replenishment time $Tz[Y]$ of the magenta toner, which is 1.6 s, remains unchanged.

[0087] That is, since the accumulation necessary replenishment time $Tz[Y]$ ($=1.6$ s) of the magenta toner at the time point when the replenishment of the yellow toner for the time of 1 s is finished is equal to or greater than the toner replenishment threshold Ta ($=1$ s), the replenishment of the magenta toner is resumed. This replenishment of the magenta toner is performed for the accumulation necessary replenishment time $Tz[M]$ of the magenta toner with the continuous replenishment allowable time Ton ($=1$ s) as an upper limit, that is, performed for the continuous replenishment allowable time Ton of 1 s. Accordingly, the accumulation necessary replenishment time $Tz[M]$ of the magenta toner becomes 0.6 s ($=1.6$ s -1 s). On the other hand, the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner, which is 1.6 s, remains unchanged.

[0088] Since the accumulation necessary replenishment time $Ts[Y]$ ($=1.6$ s) of the yellow toner at the time point when this replenishment of the magenta toner is finished is equal to or greater than the toner replenishment threshold Ta ($=1$ s), the replenishment of the yellow toner is resumed. Thereafter, the replenishment of the magenta toner and the replenishment of the yellow toner are alternately performed in a similar manner.

[0089] When the accumulation necessary replenishment time $Tz[M]$ of the magenta toner and the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner in this process are compared, the mutual difference between them ($=|Tz[M]-Tz[Y]|$) is 1 s at the maximum. The mutual difference (time) of 1 s is equivalent to 0.2 g when converted into the amount of toner, and is extremely small. This means that the replenishment of the magenta toner and the replenishment of the yellow toner are performed in a very good balance, and a good density balance is kept also on the output image.

[0090] In FIG. 6, the total time surrounded by the short broken line is 1 s, that is, the continuous replenishment allowable time Ton . In FIG. 6, the total time surrounded by the long broken line is 1.71 s, that is, the printing time Tp per sheet of paper. That is, after one of the replenishment of the magenta toner and the replenishment of the yellow toner is performed for the continuous replenishment allowable time Ton , the other of the replenishment of the magenta toner and the replenishment of the yellow toner is performed for the continuous replenishment allowable time Ton , that is, the replenishment target transitions (switches) as indicated by the arrow in FIG. 6. This transition of the replenishment target is performed asynchronously with the print state.

[0091] Here, the first replenishment apparatus 37a, that is, the selective replenishment manner of the yellow toner and the magenta toner has been described, but also the second replenishment apparatus 37b, that is, selective replenishment is performed in a similar manner also for the cyan toner and the black toner. That is, the selective replenishment is performed in an extremely good balance also for the cyan toner and the black toner, and a good density balance is kept also on the output image.

[0092] As seen from FIG. 6, each time printing in which the printing rate $Rp[Y]$ of yellow is 100% and the printing rate $Rp[M]$ of magenta is 100% is performed, each of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner increases. When each of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner excessively increases, each of the replenishment of the yellow toner and the replenishment of the magenta toner is excessively not in time, and each of the amount of the yellow toner and the amount of the magenta toner to be used for development is insufficient. As a result, an image quality defect of density reduction occurs in the output image.

[0093] In order to avoid this, when the accumulation necessary replenishment time $Tz[X]$ of the toner of any color becomes equal to or greater than a forced replenishment threshold Tb as the predetermined third threshold, the printing is interrupted, and forced replenishment of performing replenishment for the accumulation necessary replenishment time $Tz[X]$ for the toner of all colors, respectively. However, when the forced replenishment is frequently performed, printing is interrupted accordingly, and thus productivity is reduced. Therefore, it is important that the forced replenishment threshold Tb which is a criterion for determining whether to perform forced replenishment is appropriately determined. The forced replenishment threshold Tb in the present first example is, for example, 5 s. The specific value of this forced replenishment threshold Tb is an example, and is not limited thereto, but it is assumed that the forced replenishment threshold Tb is a value larger than the toner replenishment threshold Ta and the continuous replenishment allowable time Ton described above.

[0094] Furthermore, a case of continuously performing printing in which the printing rate $Rp[Y]$ of yellow is 0% and the printing rate $Rp[M]$ of magenta is 100% will be described with reference to FIG. 7. FIG. 7 illustrates an example in which the accumulation necessary replenishment time

Tz[M] of the magenta toner immediately before a job instructing execution of printing is received is 0 s, and the accumulation necessary replenishment time Tz[Y] of the yellow toner is less than the toner replenishment threshold Ta.

[0095] As illustrated in FIG. 7, when printing of the first sheet is performed, the accumulation necessary replenishment time Tz[M] of the magenta toner becomes 1.3 s. Then, replenishment of the magenta toner is performed. This replenishment of the magenta toner is performed for the accumulation necessary replenishment time Tz[M] of the magenta toner with the continuous replenishment allowable time Ton (=1 s) as an upper limit, that is, for the continuous replenishment allowable time Ton of 1 s. Accordingly, the accumulation necessary replenishment time Tz[M] of the magenta toner becomes 0.3 s (=1.3 s−1 s). On the other hand, although not illustrated, the accumulation necessary replenishment time Tz[Y] of the yellow toner remains less than the toner replenishment threshold Ta as described above.

[0096] After the replenishment of the magenta toner is stopped for the replenishment stop time Toff (=0.1 s), the replenishment of the magenta toner is resumed. This replenishment of the magenta toner is performed until the accumulation necessary replenishment time Tz[M] of the magenta toner becomes 0, that is, performed for 0.3 s. Thereafter, the replenishment of the magenta toner is stopped until the printing of the second sheet is started, that is, until a time of 0.31 s (=1.71 s−1 s−0.1 s−0.3 s) elapses (this stop time of 0.1 s is provided when the accumulation necessary replenishment time Tz[Y] of the partner yellow toner is less than the toner replenishment threshold Ta, that is, when it is not necessary to switch the rotation direction of the first replenishment motor **110a** from the other direction R2 to the one direction R1. In this way, the determination processing of a control program of the first replenishment motor **110a** can be simplified).

[0097] When printing of the second sheet is performed, the accumulation necessary replenishment time Tz[M] of the magenta toner becomes 1.3 s again. Thereafter, replenishment of the magenta toner is performed in a similar manner.

[0098] As seen from FIG. 7, in a case of continuously performing printing in which the printing rate Rp[Y] of yellow is 0% and the printing rate Rp[M] of magenta is 100%, the accumulation necessary replenishment time Tz[M] of the magenta toner does not become greater than 1.3 s, that is, does not excessively increase. Therefore, the above-described forced replenishment is not performed, and productivity is kept.

[0099] In FIG. 7, the time surrounded by the short broken line is a replenishment time per one time of the magenta toner, that is, the continuous replenishment allowable time Ton (=1 s). In FIG. 7, the total time surrounded by the long broken line is 1.71 s, that is, the printing time Tp per sheet of paper.

[0100] FIG. 8 is a block diagram illustrating an electrical configuration of the image forming apparatus **10**. As illustrated in FIG. 8, the image forming apparatus **10** includes a controller **200**. The image reader **12**, the automatic document feeder **18**, the image former **26**, and the sheet feeder **64** are connected to the controller **200** via a bus **202**. In addition, an operation unit **204**, an auxiliary storage **206**, a communicator **208**, and the like are connected to the controller **200** via the bus **202**. The image forming apparatus **10** includes various elements other than them. However, illustration and description of elements not directly related to the gist of the present disclosure are omitted here. The image reader **12**, the automatic document feeder **18**, the image former **26**, and the sheet feeder **64** are configured as described above. In particular, the image former **26** includes the first replenishment motor **110a** and the second replenishment motor **110b**.

[0101] The controller **200** is a controller that controls the entire image forming apparatus **10**. Therefore, the controller **200** includes a computer as a control execution means, for example, a CPU **200a**. In addition, the controller **200** includes a main storage **200b** as a main storage that can be directly accessed by the CPU **200a**. The main storage **200b** includes, for example, a ROM and a RAM not illustrated. The ROM stores a control program (firmware) that controls the operation of the CPU **200a**. This control program includes a toner replenishment control program described

later. The RAM constitutes a work area and a buffer area when the CPU **200a** executes processing according to the control program.

[0102] The operation unit **204** has a display with a touchscreen not illustrated. The display with a touchscreen is a component integrally combining a touchscreen as an example of an operation receptor that can receive an operation by a user not illustrated and a display as an example of a display that displays various types of information. The operation unit **204** includes an appropriate light emitter such as an LED not illustrated and an appropriate hardware switch such as a push button not illustrated, in addition to the display with a touchscreen.

[0103] The auxiliary storage **206** is an example of an auxiliary storage. That is, the auxiliary storage **206** appropriately stores various data such as the read image data and the job described above. This auxiliary storage **206** includes, for example, a hard disk drive not illustrated. In addition, the auxiliary storage **206** may include a rewritable nonvolatile memory such as a flash memory.

[0104] The communicator **208** is an example of a communicator. That is, the communicator **208** performs bidirectional communication processing via a LAN line not illustrated. The communicator **208** may be connected to the LAN line by wire or wirelessly. The communicator **208** also performs bidirectional communication processing via a public switched telephone network not illustrated.

[0105] As described above, according to the present first example, in particular, according to the first replenishment apparatus **37a**, when at least one of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner becomes equal to or greater than the toner replenishment threshold Ta , replenishment of the one toner α related to the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner or the accumulation necessary replenishment time $Tz[M]$ of the magenta toner that becomes equal to or greater than the toner replenishment threshold Ta is performed. This replenishment of the one toner α is performed for the accumulation necessary replenishment time $Tz[a]$ of the one toner α with the continuous replenishment allowable time Ton as an upper limit. Thereafter, according to the accumulation necessary replenishment time $Tz[\beta]$ of the other toner β and the current accumulation necessary replenishment time $Tz[a]$ of the one toner α , the replenishment of the other toner β is performed, the replenishment of the one toner α is resumed, or the replenishment of both the toners α and β is brought into a stop state. The toner replenishment is performed in a similar manner also in the second replenishment apparatus **37b**.

[0106] In order to implement toner replenishment in such a manner, the CPU **200a** executes a toner replenishment control task according to the above-described toner replenishment control program. FIG. **9** shows the flow of this toner replenishment control task. The toner replenishment control task is executed in response to reception of a job instructing execution of printing. The job mentioned here includes jobs in the printer function, the copy function, and the fax function (fax reception function).

[0107] According to this toner replenishment control task, the CPU **200a** first performs initial check in step **S1**. In this initial check, with reference to history information of the previous job, the CPU **200a** checks the accumulation necessary replenishment time $Tz[X]$ of each toner at the present time point (at the start time point of the toner replenishment control task). The history information of job is stored in, for example, the auxiliary storage **206**. The CPU **200a** advances the processing to step **S3**.

[0108] In step **S3**, the CPU **200a** sets a variable n representing the number of printed sheets to a value of 1 which is an initial value of the variable n . In subsequent step **S5**, the CPU **200a** develops (analyzes) image data to be used for printing the n th sheet. Furthermore, in subsequent step **S7**, the CPU **200a** derives the printing rate $Rx[X]$ related to each toner based on the image data developed in step **S5**. In subsequent step **S9**, the CPU **200a** starts printing of the n th sheet, and then advances the processing to step **S11**. The printing started in step **S9** is performed by a print task different

from the toner replenishment control task. Detailed description of this print task including illustration thereof will be omitted.

[0109] In step **S11**, the CPU **200a** derives the necessary replenishment time $Ts[X]$ ($=Qc[X]/Vs[X]$) for each toner. In this deriving of the necessary replenishment time $Ts[X]$, the consumption amount $Qc[X]$ of each toner is used, and this consumption amount $Qc[X]$ is obtained based on a relational expression between the printing rate $Rp[X]$ and the consumption amount $Qc[X]$ related to each toner, for example. Alternatively, the consumption amount $Qc[X]$ may be obtained based on a lookup table in which the relationship between the printing rate $Rp[X]$ and the consumption amount $Qc[X]$ related to each toner is summarized. After the execution of step **S11**, the CPU **200a** advances the processing to step **S13**.

[0110] In step **S13**, the CPU **200a** adds the necessary replenishment time $Ts[X]$ of each toner derived in step **S11** to the accumulation necessary replenishment time $Tz[X]$ of each toner at the present time point, and sets the added value as a new accumulation necessary replenishment time $Tz[X]$, that is, updates the accumulation necessary replenishment time $Tz[X]$. The CPU **200a** advances the processing to step **S15**.

[0111] In step **S15**, the CPU **200a** compares the accumulation necessary replenishment time $Tz[X]$ of each toner with the forced replenishment threshold Tb , and determines whether the accumulation necessary replenishment time $Tz[X]$ of each toner is less than the forced replenishment threshold Tb . Here, if the accumulation necessary replenishment time $Tz[X]$ of any toner is equal to or greater than the forced replenishment threshold Tb (**S15: NO**), the CPU **200a** advances the processing to step **S17**. On the other hand, if the accumulation necessary replenishment time $Tz[X]$ of all the toners is less than the forced replenishment threshold Tb (**S15: YES**), the CPU **200a** advances the processing to step **S19** described later.

[0112] In step **S17**, the CPU **200a** starts the above-described forced replenishment and ends the toner replenishment control task. The forced replenishment is performed by a forced replenishment task different from the toner replenishment control task. Detailed description of this forced replenishment task including illustration thereof will be omitted. When the forced replenishment is started, printing is interrupted. When the forced replenishment ends, printing is resumed.

[0113] On the other hand, when advancing the processing from step **S15** to step **S19**, the CPU **200a** compares the accumulation necessary replenishment time $Tz[X]$ of each toner with the toner replenishment threshold Ta in step **S19**, and determines whether the accumulation necessary replenishment time $Tz[X]$ of each toner is less than the toner replenishment threshold Ta . Here, if the accumulation necessary replenishment time $Tz[X]$ of any toner is equal to or greater than the toner replenishment threshold Ta (**S19: NO**), the CPU **200a** advances the processing to step **S21**. On the other hand, if the accumulation necessary replenishment time $Tz[X]$ of all the toners is less than the toner replenishment threshold Ta (**S19: YES**), the CPU **200a** advances the processing to step **S23** described later.

[0114] In step **S21**, the CPU **200a** executes toner replenishment processing that performs toner replenishment. This toner replenishment processing will be described in detail later. After finishing this toner replenishment processing, the CPU **200a** advances the processing to step **S23**.

[0115] In step **S23**, the CPU **200a** determines whether all the printing according to the job is finished. Here, if all the printing according to the job is finished (**S23: YES**), the CPU **200a** ends the toner replenishment control task. On the other hand, if all the printing according to the job is not finished yet, the CPU **200a** advances the processing to step **S25**. In step **S25**, the CPU **200a** increments the value of the variable n representing the number of printed sheets, and then returns the processing to step **S5**.

[0116] Next, the toner replenishment processing will be described with reference to FIGS. **10** to **12**. Note that FIGS. **10** to **12** illustrate the flow of the toner replenishment processing for the first replenishment apparatus **37a**, but the toner replenishment processing is also performed for the second replenishment apparatus **37b** in the same flow.

[0117] According to this toner replenishment processing, first, in step **S101**, the CPU **200a** resets and starts a timer not illustrated that measures the printing time T_p (strictly speaking, the printing time T_p in which the processing time of the toner replenishment control task including the toner replenishment processing is taken into consideration (subtracted)) per sheet. This timer is a software timer configured by the CPU **200a**, for example, but may be a hardware timer configured by a hardware element such as a real-time clock (RTC). After the execution of this step **S101**, the CPU **200a** advances the processing to step **S103**.

[0118] In step **S103**, the CPU **200a** compares the accumulation necessary replenishment time $T_z[Y]$ of the yellow toner with the accumulation necessary replenishment time $T_z[M]$ of the magenta toner, and determines whether both are equivalent to each other. Here, if both are equivalent to each other (**S103: YES**), the CPU **200a** advances the processing to step **S105**. On the other hand, if both are not equivalent to each other (**S103: NO**), the CPU **200a** advances the processing to step **S113** described later.

[0119] In step **S105**, the CPU **200a** compares the toner density detection value $T/D[Y]$ of the yellow toner with the toner density detection value $T/D[M]$ of the magenta toner, and determines whether both are equivalent to each other. Here, if both are equivalent to each other (**S105: YES**), the CPU **200a** advances the processing to step **S107**. On the other hand, if both are not equivalent to each other (**S105: NO**), the CPU **200a** advances the processing to step **S109** described later.

[0120] In step **S107**, the CPU **200a** specifies the yellow toner as the replenishment target α , that is, the yellow toner as the one toner α described above. In addition, the CPU **200a** specifies the magenta toner as the replenishment target β , that is, the magenta toner as the other toner β described above. The CPU **200a** advances the processing to step **S115** described later.

[0121] On the other hand, when advancing the processing from step **S105** to step **S109** described above, the CPU **200a** determines whether the toner density detection value $T/D[Y]$ of the yellow toner is less than the toner density detection value $T/D[M]$ of the magenta toner in step **S109**. Here, if the toner density detection value $T/D[Y]$ of the yellow toner is less than the toner density detection value $T/D[M]$ of the magenta toner (**S109: YES**), the CPU **200a** advances the processing to step **S107**. On the other hand, if the toner density detection value $T/D[Y]$ of the yellow toner is larger than the toner density detection value $T/D[M]$ of the magenta toner (**S109: NO**), the CPU **200a** advances the processing to step **S111** described later.

[0122] In step **S111**, the CPU **200a** specifies the yellow toner as the replenishment target β , that is, the yellow toner as the other toner β . In addition, the CPU **200a** specifies the magenta toner as the replenishment target α , that is, specifies the magenta toner as the one toner α . The CPU **200a** advances the processing to step **S115** described later.

[0123] Furthermore, when advancing the processing from step **S103** to step **S113** described above, the CPU **200a** determines whether the accumulation necessary replenishment time $T_z[Y]$ of the yellow toner is longer than the accumulation necessary replenishment time $T_z[M]$ of the magenta toner in step **S113**. Here, if the accumulation necessary replenishment time $T_z[Y]$ of the yellow toner is longer than the accumulation necessary replenishment time $T_z[M]$ of the magenta toner (**S113: YES**), the CPU **200a** advances the processing to step **S107**. On the other hand, if the accumulation necessary replenishment time $T_z[Y]$ of the yellow toner is shorter than the accumulation necessary replenishment time $T_z[M]$ of the magenta toner (**S113: NO**), the CPU **200a** advances the processing to step **S111**.

[0124] In step **S115**, the CPU **200a** resets a counter that counts the replenishment time T_d of the toner being replenishment target, that is, sets the count value to 0 as an initial value. The CPU **200a** advances the processing to step **S117**.

[0125] In step **S117**, the CPU **200a** compares the count value of the counter that counts the replenishment time T_d with the continuous replenishment allowable time T_{on} , and determines whether the replenishment time T_d is less than the continuous replenishment allowable time T_{on} . Here, if the replenishment time T_d is less than the continuous replenishment allowable time T_{on}

(S117: YES), the CPU **200a** advances the processing to step S119. On the other hand, if the replenishment time T_d is equal to or greater than the continuous replenishment allowable time T_{on} (S117: NO), the CPU **200a** advances the processing to step S129 described later.

[0126] In step S119, the CPU **200a** drives the first replenishment motor **110a** so that replenishment of the one toner α is performed for a predetermined replenishment unit time ΔT . The replenishment unit time ΔT mentioned here is a time sufficiently shorter than the continuous replenishment allowable time T_{on} ($=1$ s), and is, for example, 0.1 s. After the execution of step S119, the CPU **200a** advances the processing to step S121.

[0127] In step S121, the CPU **200a** subtracts the replenishment unit time ΔT from the current accumulation necessary replenishment time $T_z[a]$ of the one toner α , and sets the subtracted value as a new accumulation necessary replenishment time $T_z[a]$, that is, updates the accumulation necessary replenishment time $T_z[a]$. The CPU **200a** advances the processing to step S123.

[0128] In step S123, the CPU **200a** subtracts the replenishment unit time ΔT from the count value of the counter that counts the replenishment time T_d described above, and sets the subtracted value as a new count value, that is, updates the count value of the replenishment time T_d . The CPU **200a** advances the processing to step S125.

[0129] In step S125, the CPU **200a** determines whether the measurement time by the timer described above has not yet passed the printing time T_p per sheet, that is, whether the printing time T_p has not yet elapsed. Here, if the printing time T_p has not yet elapsed (S125: YES), the CPU **200a** advances the processing to step S127. On the other hand, if the printing time T_p has elapsed (S125: NO), the CPU **200a** ends the toner replenishment processing.

[0130] In step S127, the CPU **200a** determines whether the current accumulation necessary replenishment time $T_z[a]$ of the one toner α is 0 or less. Here, if the current accumulation necessary replenishment time $T_z[a]$ of the one toner α is not 0 or less (S127: NO), the CPU **200a** returns the processing to step S117. On the other hand, if the current accumulation necessary replenishment time $T_z[a]$ of the one toner α is 0 or less (S127: YES), the CPU **200a** advances the processing to step S129.

[0131] In step S129, the CPU **200a** compares the accumulation necessary replenishment time $T_z[\beta]$ of the other toner β with the toner replenishment threshold T_a , and determines whether the accumulation necessary replenishment time $T_z[\beta]$ is equal to or greater than the toner replenishment threshold T_a . Here, if the accumulation necessary replenishment time $T_z[\beta]$ is less than the toner replenishment threshold T_a (S129: NO), the CPU **200a** advances the processing to step S131. In step S131, the CPU **200a** waits for the replenishment stop time T_{off} , and then returns the processing to step S115. On the other hand, in step S129, if the accumulation necessary replenishment time $T_z[\beta]$ is equal to or greater than the toner replenishment threshold T_a (S129: YES), the CPU **200a** advances the processing to step S133.

[0132] In step S133, the CPU **200a** resets the above-described counter that counts the replenishment time T_d of the toner being replenishment target, and then advances the processing to step S135. In step S135, the CPU **200a** compares the count value of the replenishment time T_d by the counter with the continuous replenishment allowable time T_{on} , and determines whether the replenishment time T_d is less than the continuous replenishment allowable time T_{on} . Here, if the replenishment time T_d is equal to or greater than the continuous replenishment allowable time T_{on} (S135: NO), the CPU **200a** returns the processing to step S115. On the other hand, if the replenishment time T_d is less than the continuous replenishment allowable time T_{on} (S135: YES), the CPU **200a** advances the processing to step S137.

[0133] In step S137, the CPU **200a** drives the first replenishment motor **110a** so that the replenishment of the other toner β is performed for the above-described replenishment unit time ΔT . Thereafter, the CPU **200a** advances the processing to step S139.

[0134] In step S139, the CPU **200a** subtracts the replenishment unit time ΔT from the current accumulation necessary replenishment time $T_z[\beta]$ of the other toner, and sets the subtracted value

as a new accumulation necessary replenishment time $Tz[\beta]$, that is, updates the accumulation necessary replenishment time $Tz[\beta]$. The CPU **200a** advances the processing to step **S141**.

[0135] In step **S141**, the CPU **200a** subtracts the replenishment unit time ΔT from the count value of the replenishment time Td by the counter described above, and sets the subtracted value as a new count value, that is, updates the count value of the replenishment time Td . The CPU **200a** advances the processing to step **S143**.

[0136] In step **S143**, the CPU **200a** determines whether the measurement time by the timer described above has not yet passed the printing time Tp per sheet, that is, whether the printing time Tp has not yet elapsed. Here, if the printing time Tp has not yet elapsed (**S143: YES**), the CPU **200a** advances the processing to step **S145**. On the other hand, if the printing time Tp has elapsed (**S143: NO**), the CPU **200a** ends the toner replenishment processing.

[0137] In step **S145**, the CPU **200a** determines whether the current accumulation necessary replenishment time $Tz[\beta]$ of the other toner β is 0 or less. Here, if the current accumulation necessary replenishment time $Tz[\beta]$ of the other toner β is not 0 or less (**S145: NO**), the CPU **200a** returns the processing to step **S135**. On the other hand, if the current accumulation necessary replenishment time $Tz[\beta]$ of the other toner β is 0 or less (**S147: YES**), the CPU **200a** returns the processing to step **S115**.

[0138] Thus, according to the present first example, in particular, according to the first replenishment apparatus **37a**, when at least one of the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner and the accumulation necessary replenishment time $Tz[M]$ of the magenta toner becomes equal to or greater than the toner replenishment threshold Ta , replenishment of the one toner α related to the accumulation necessary replenishment time $Tz[Y]$ of the yellow toner or the accumulation necessary replenishment time $Tz[M]$ of the magenta toner that becomes equal to or greater than the toner replenishment threshold Ta is performed. This replenishment of the one toner α is performed for the accumulation necessary replenishment time $Tz[a]$ of the one toner α with the continuous replenishment allowable time Ton as an upper limit. Thereafter, according to the accumulation necessary replenishment time $Tz[\beta]$ of the other toner β and the current accumulation necessary replenishment time $Tz[a]$ of the one toner α , the replenishment of the other toner β is performed, the replenishment of the one toner α is resumed, or the replenishment of both the toners a and β is brought into a stop state. The toner replenishment is performed in a similar manner also in the second replenishment apparatus **37b**.

[0139] That is, according to the present first example, unlike the technique disclosed in Patent Document 1 described above, that is, unlike the technique disclosed in Patent Document 1 requiring performing of complicated processing such as obtaining a ratio between the amount of the first developer to be supplied to the first development apparatus and the amount of the second developer to be supplied to the second development apparatus and allocating the first drive time and the second drive time according to the ratio, it is possible to perform replenishment of two types of toners a and β with one drive source in a good balance without performing such complicated processing, in other words, by an extremely simple and uncomplicated manner. This is extremely advantageous for the configuration as in the present first example where the cost of the entire image forming apparatus **10** is reduced by performing replenishment of the two types of toners a and β by one drive source.

[0140] In the present first example, in particular, the CPU **200a** that executes step **S11** of the toner replenishment control task, that is, the CPU **200a** that derives the necessary replenishment time $Ts[X]$ for each toner is an example of a derivator according to the present disclosure. The CPU **200a** that executes step **S13** of the toner replenishment control task, that is, the CPU **200a** that updates the accumulation necessary replenishment time $Tz[X]$ of each toner is an example of the accumulator according to the present disclosure. Furthermore, the first replenishment motor **110a** and the second replenishment motor **110b** which are drive sources are driven under the control of the CPU **200a**, and the CPU **200a** responsible for controlling the first replenishment motor **110a**

and the second replenishment motor **110b** is an example of a drive controller according to the present disclosure.

Second Example

[0141] Next, the second example of the present disclosure will be described.

[0142] In the present first example described above, the forced replenishment threshold T_b common to each other is used for each toner, and the forced replenishment threshold T_b is assumed to be constant. However, in the present second example, an individual forced replenishment threshold $T_b[X]$ is used for each toner. In addition, in the present second example, the forced replenishment threshold $T_b[X]$ for each toner is set (changed) based on a mean printing rate $R_a[X]$ which is the mean value of the printing rates $R_p[X]$ for each toner of the most recent N sheets.

[0143] That is, as described above, when the forced replenishment is frequently performed, printing is interrupted accordingly, and thus productivity is reduced. On the other hand, when the timing at which the forced replenishment is performed is late, there is a possibility that an image quality defect of density reduction occurs in the output image. When the replenishment amount of toner by forced replenishment is excessively large, the replenished toner is not sufficiently charged, that is, charging is insufficient, charging unevenness occurs, and another image quality defect of a fogging phenomenon in which a toner image is formed in a part where the toner image is not originally formed on the sheet may occur. Furthermore, when the replenishment amount of toner by forced replenishment is excessively large, the replenished toner may be scattered to the outside of (the housing of) the development apparatus **50**. Therefore, it is important that the forced replenishment is performed at an appropriate timing.

[0144] Therefore, in the present second example, the individual forced replenishment threshold $T_b[X]$ is used for each toner as described above, and the forced replenishment threshold $T_b[X]$ for each toner is set based on the mean printing rate $R_a[X]$ which is the mean value of the printing rates $R_p[X]$ for each toner of the most recent N sheets. For this purpose, a forced replenishment threshold table **300** as illustrated in FIG. **13** is provided. The forced replenishment threshold table **300** is incorporated in the toner replenishment control program.

[0145] According to this forced replenishment threshold table **300**, when the mean printing rate $R_a[X]$ of the most recent N sheets is 5% or less, for example, a value of 5.0 s is set as the forced replenishment threshold $T_b[X]$ for each toner. When the mean printing rate $R_a[X]$ is over 5% and is 10% or less, a value of 4.5 s is set as the forced replenishment threshold $T_b[X]$, and when the mean printing rate $R_a[X]$ is over 10% and is 25% or less, a value of 4.0 s is set as the forced replenishment threshold $T_b[X]$. Furthermore, when the mean printing rate $R_a[X]$ is over 25% and 40% or less, a time of 3.5 s is set as the forced replenishment threshold $T_b[X]$, and when the mean printing rate $R_a[X]$ is over 40% and 60% or less, a value of 3.0 s is set as the forced replenishment threshold $T_b[X]$. When the mean printing rate $R_a[X]$ is over 60% and is 80% or less, a value of 2.0 s is set as the forced replenishment threshold $T_b[X]$, and when the mean printing rate $R_a[X]$ is over 80%, a value of 2.0 s is set as the forced replenishment threshold $T_b[X]$.

[0146] That is, as the mean printing rate $R_a[X]$ of the most recent N sheets is lower, a longer time is set as the forced replenishment threshold $T_b[X]$, and as the mean printing rate $R_a[X]$ of the most recent N sheets is higher, a shorter time is set as the forced replenishment threshold $T_b[X]$. In short, an appropriate forced replenishment threshold $T_b[X]$ corresponding to the mean printing rate $R_a[X]$ of the most recent N sheets is set for each toner. Thus, forced replenishment is performed at an appropriate timing according to the mean printing rate $R_a[X]$ of the most recent N sheets for each toner.

[0147] The value of N is, for example, 5. When the value of N is excessively large, the timing at which the forced replenishment is performed is delayed and the above-described image quality defect may occur. On the other hand, when the value of N is excessively small, the forced replenishment is frequently performed and thus productivity may be reduced. The value of N being 5 was derived in view of them. However, the value of N is not limited to 5 and may be arbitrarily

changeable.

[0148] Also in the present second example, the above-described toner replenishment control task is executed. In particular, in the initial check in step **S1**, in addition to checking the accumulation necessary replenishment time $Tz[X]$ of each toner at the present time point, the mean printing rate $Ra[X]$ of the most recent N sheets for each toner is checked. Then, at an appropriate stage after step **S7** and before step **S15**, the mean printing rate $Ra[X]$ of the most recent N sheets is calculated again, that is, updated, and the forced replenishment threshold table **300** is referred to, whereby the forced replenishment threshold $Tb[X]$ corresponding to the updated mean printing rate $Ra[X]$ is specified. Then, the specified forced replenishment threshold $Tb[X]$ is applied as a comparison target of the accumulation necessary replenishment time $Tz[X]$ in step **S15**.

[0149] As described above, according to the present second example, the forced replenishment is performed at an appropriate timing according to the mean printing rate $Ra[X]$ of the most recent N sheets for each toner. Thus, it is possible to prevent the occurrence of image quality defects such as density reduction of an output image and a fogging phenomenon while suppressing the frequency of performing forced replenishment and keeping high productivity, and further, it is possible to prevent the replenished toner from scattering to the outside of (the housing of) the development apparatus **50**.

Third Example

[0150] Next, the third example of the present disclosure will be described.

[0151] The present third example is based on the present first example. In the present third example, for each toner, a threshold coefficient $\gamma[X]$ corresponding to the toner density detection value $T/D[X]$ by the toner density sensor **50b** described above is multiplied by the forced replenishment threshold Tb , and the multiplication value $\gamma[X] \cdot Tb$ is applied as a comparison target of the accumulation necessary replenishment time $Tz[X]$. For this purpose, a threshold coefficient table **400** as illustrated in FIG. **14** is provided. The threshold coefficient table **400** is incorporated in the toner replenishment control program.

[0152] According to this threshold coefficient table **400**, for each toner, for example, when the toner density detection value $T/D[X]$ is 5.0 wt % or less, a value of 0.75 is multiplied by the forced replenishment threshold Tb as the threshold coefficient $\gamma[X]$. In this case, since the multiplication value $\gamma[X] \cdot Tb$ that is a comparison target of the accumulation necessary replenishment time $Tz[X]$ is a value smaller than the forced replenishment threshold Tb , the forced replenishment of the toner is easily performed. When the toner density detection value $T/D[X]$ is over 5.0 wt % and 5.5 wt % or less, a value of 1.00 as the threshold coefficient $\gamma[X]$ is multiplied by the forced replenishment threshold Tb . In this case, the multiplication value $\gamma[X] \cdot Tb$ that is a comparison target of the accumulation necessary replenishment time $Tz[X]$ is equivalent to the forced replenishment threshold Tb . Furthermore, when the toner density detection value $T/D[X]$ is over 5.5 wt % and 6.5 wt % or less, a value of 1.25 as the threshold coefficient $\gamma[X]$ is multiplied by the forced replenishment threshold Tb . In this case, since the multiplication value $\gamma[X] \cdot Tb$ that is a comparison target of the accumulation necessary replenishment time $Tz[X]$ is a value larger than the forced replenishment threshold Tb , the forced replenishment of toner is hardly performed. An appropriate value of the toner density T/D is, for example, 6.0 wt %.

[0153] When the toner density detection value $T/D[X]$ is over 6.5 wt % and 7.0 wt % or less, a value of 1.00 as the threshold coefficient $\gamma[X]$ is multiplied by the forced replenishment threshold Tb . In this case, the multiplication value $\gamma[X] \cdot Tb$ that is a comparison target of the accumulation necessary replenishment time $Tz[X]$ is equivalent to the forced replenishment threshold Tb . When the toner density detection value $T/D[X]$ is over 7.0 wt %, a value of 0.75 as the threshold coefficient $\gamma[X]$ is multiplied by the forced replenishment threshold Tb . In this case, since the multiplication value $\gamma[X] \cdot Tb$ that is a comparison target of the accumulation necessary replenishment time $Tz[X]$ is a value smaller than the forced replenishment threshold Tb , the forced replenishment of toner is easily performed.

[0154] That is, when the toner density detection value $T/D[X]$ is 5.0 wt % or less, that is, when the toner density T/D is excessively lower than an appropriate value thereof, the forced replenishment threshold T_b is corrected so that forced replenishment the toner is easily performed. When the toner density T/D is excessively low, there is a possibility that the density of the output image decreases and the stirring unevenness of the developer occurs. Therefore, in order to avoid these disadvantages, the forced replenishment threshold T_b is corrected so that the forced replenishment the toner is easily performed.

[0155] Also when the toner density detection value $T/D[X]$ is over 7.0 wt %, that is, when the toner density T/D is excessively higher than an appropriate value thereof, the forced replenishment threshold T_b is corrected so that the forced replenishment of toner is easily performed. When the toner density T/D is excessively high, there is a possibility that the toner in the development apparatus **50** (the developer accommodation chamber **50a**) scatters to the outside of the development apparatus **50** or the above-described fogging phenomenon occurs. Therefore, in order to avoid these disadvantages, the forced replenishment threshold T_b is corrected so that the forced replenishment of the toner is easily performed.

[0156] When the toner density detection value $T/D[X]$ is over 5.5 wt % and 6.5 wt % or less, that is, when the toner density T/D is approximately an appropriate value, the forced replenishment threshold T_b is corrected so that the forced replenishment of the toner is hardly performed. When the toner density T/D is approximately an appropriate value, there is almost no possibility that disadvantages such as a case where the toner density T/D is excessively high or excessively low occurs. Therefore, in order to further improve productivity, the forced replenishment threshold T_b is corrected so that the forced replenishment of the toner is hardly performed.

[0157] Also in the present third example, the above-described toner replenishment control task is executed. In particular, in the initial check in step **S1**, in addition to checking the accumulation necessary replenishment time $T_z[X]$ of each toner at the present time point, the toner density detection value $T/D[X]$ for each toner is checked. In step **S1** or at an appropriate stage after step **S1** and before step **S15**, the threshold coefficient $\gamma[X]$ corresponding to the toner density detection value $T/D[X]$ for each toner is specified with reference to the threshold coefficient table **400**. Then, the specified threshold coefficient $\gamma[X]$ is multiplied by the forced replenishment threshold T_b , and the multiplication value $\gamma[X] \cdot T_b$ is applied as a comparison target of the accumulation necessary replenishment time $T_z[X]$ in step **S15**.

[0158] As described above, according to the present third example, for each toner, the multiplication value $\gamma[X] \cdot T_b$ of the threshold coefficient $\gamma[X]$ corresponding to the toner density detection value $T/D[X]$ and the forced replenishment threshold T_b is a comparison target with the accumulation necessary replenishment time $T_z[X]$. Thus, it is possible to further improve productivity while suppressing occurrence of disadvantages when the toner density T/D is excessively high or excessively low for each toner.

[0159] The present third example is based on the present first example, but may be based on the present second example. In this case, for each toner, a multiplication value $\gamma[X] \cdot T_b[X]$ of the threshold coefficient $\gamma[X]$ and the forced replenishment threshold $T_b[X]$ is a comparison target with the accumulation necessary replenishment time $T_z[X]$.

Other Application Examples

[0160] Each of the above examples is a specific example of the present disclosure, and does not limit the technical scope of the present disclosure. The present disclosure can be applied to aspects other than these examples.

[0161] For example, the developer is not limited to a two-component system, and may be a one-component system developer not containing a carrier. However, the third example is based on a two-component developer.

[0162] In each example, the case where the size of the sheet is the A4 size has been described, but the size of the sheet is not limited to the A4 size.

[0163] Furthermore, in each example, the image forming apparatus **10** employing the tandem-type color image former **26** has been described as an example, but the present disclosure can also be applied to an image forming apparatus employing a rotary-type color image former.

[0164] In addition, the image forming apparatus **10** in each example is a multifunction peripheral. However, the present disclosure can also be applied to an image forming apparatus other than the multifunction peripheral, such as a dedicated printing machine, a dedicated copy machine, or a dedicated fax machine.

[0165] The present disclosure can be provided not only in the form of an apparatus called an image forming apparatus, but also in the form of a method called a toner replenishment control method in an image forming apparatus.

Claims

1. An image forming apparatus that performs image formation processing by an electrophotographic method, the image forming apparatus comprising: a first accommodation in which first toner is accommodated; a first replenishment member that replenishes a first development unit with the first toner accommodated in the first accommodation; a second accommodation in which second toner is accommodated; a second replenishment member that replenishes a second development unit with the second toner accommodated in the second accommodation; a drive source that selectively drives the first replenishment member and the second replenishment member; a drive controller that controls the drive source; a derivator that derives, based on image data to be subjected to the image formation processing, a first necessary replenishment time, which is a replenishment time of the first toner necessary for compensating for consumption of the first toner consumed by the first development unit, and a second necessary replenishment time, which is a replenishment time of the second toner necessary for compensating for consumption of the second toner consumed by the second development unit; and an accumulator that accumulates a shortage of an actual replenishment time of the first toner by the first replenishment member with respect to the first necessary replenishment time every time the image formation processing is performed and obtains a first accumulation time, which is an accumulation value of the shortage, and that accumulates a shortage of an actual replenishment time of the second toner by the second replenishment member with respect to the second necessary replenishment time every time the image formation processing is performed and obtains a second accumulation time, which is an accumulation value of the shortage, wherein when at least one of the first accumulation time and the second accumulation time is equal to or greater than a predetermined first threshold, the drive controller specifies the first accumulation time or the second accumulation time that is equal to or greater than the first threshold as one accumulation time, specifies the first replenishment member or the second replenishment member related to the one accumulation time as one replenishment member, drives the one replenishment member for the one accumulation time with a predetermined second threshold as an upper limit, and then stops drive of the one replenishment member, and further, according to another accumulation time, which is not the one accumulation time, of the first accumulation time and the second accumulation time and the one accumulation time, drives another replenishment member, which is not the one replenishment member, of the first replenishment member and the second replenishment member for the other accumulation time with the second threshold as an upper limit, or drives the one replenishment member for the one accumulation time with the second threshold as an upper limit, or controls the drive source so as to bring the one replenishment member and the other replenishment member into a stop state without driving both of the one replenishment member and the other replenishment member.

2. The image forming apparatus according to claim 1, wherein the drive controller specifies, as the one accumulation time, a longer one of the first accumulation time and the second accumulation

time when both the first accumulation time and the second accumulation time are equal to or greater than the first threshold.

3. The image forming apparatus according to claim 2 further comprising: a first toner density detector that detects a density of the first toner contained in a developer in the first development unit; and a second toner density detector that detects a density of the second toner contained in a developer in the second development unit, wherein when both the first accumulation time and the second accumulation time are equal to or greater than the first threshold and when the first accumulation time and the second accumulation time have a length equal to each other, the drive controller specifies, as the one accumulation time, the first accumulation time or the second accumulation time corresponding to a lower one of a density detection value by the first toner density detector and a density detection value by the second toner density detector.

4. The image forming apparatus according to claim 1, wherein the second threshold is a value equal to or greater than the first necessary replenishment time at a printing rate related to the first toner based on the image data is 50% and equal to or less than the first necessary replenishment time at the printing rate is 90%.

5. The image forming apparatus according to claim 1, wherein when at least one of the first accumulation time and the second accumulation time is equal to or greater than a predetermined third threshold, the drive controller controls the drive source so as to interrupt the image formation processing, drives the first replenishment member for the first accumulation time, and drive the second replenishment member for the second accumulation time.

6. A toner replenishment control method in an image forming apparatus that performs image formation processing by an electrophotographic method, the image forming apparatus including a first accommodation in which first toner is accommodated, a first replenishment member that replenishes a first development unit with the first toner accommodated in the first accommodation, a second accommodation in which second toner is accommodated, a second replenishment member that replenishes a second development unit with the second toner accommodated in the second accommodation, and a drive source that selectively drives the first replenishment member and the second replenishment member, the toner replenishment control method comprising: a drive control step of controlling the drive source, a derivation step of deriving, based on image data to be subjected to the image formation processing, a first necessary replenishment time, which is a replenishment time of the first toner necessary for compensating for consumption of the first toner consumed by the first development unit, and a second necessary replenishment time, which is a replenishment time of the second toner necessary for compensating for consumption of the second toner consumed by the second development unit, and an accumulation step of accumulating a shortage of an actual replenishment time of the first toner by the first replenishment member with respect to the first necessary replenishment time every time the image formation processing is performed and obtaining a first accumulation time, which is an accumulation value of the shortage, and of accumulating a shortage of an actual replenishment time of the second toner by the second replenishment member with respect to the second necessary replenishment time every time the image formation processing is performed and obtaining a second accumulation time which is an accumulation value of the shortage, wherein the drive control step includes, when at least one of the first accumulation time and the second accumulation time is equal to or greater than a predetermined first threshold, specifying the first accumulation time or the second accumulation time that is equal to or greater than the first threshold as one accumulation time, and specifying the first replenishment member or the second replenishment member related to the one accumulation time as one replenishment member, then driving the one replenishment member for the one accumulation time with a predetermined second threshold as an upper limit, and then stopping drive of the one replenishment member, and further, according to another accumulation time, which is not the one accumulation time, of the first accumulation time and the second accumulation time and the one accumulation time, driving another replenishment member, which is not the one

replenishment member, of the first replenishment member and the second replenishment member for the other accumulation time with the second threshold as an upper limit, or driving the one replenishment member for the one accumulation time with the second threshold as an upper limit, or controlling the drive source so as to bring the one replenishment member and the other replenishment member into a stop state without driving both of the one replenishment member and the other replenishment member.
